# Chapter 3 — Electromagnetic induction

I ⟶ Magnetic effect (chp ②) ⟵ (chp ③)
- $B_{wire}$
- $B_{coil}$
- $B_{sol.}$

I ⟷ Mag. field $\phi_m$

Variation (تغيير: قرب/بعد، زود/نقص، لف) ⟿ Induction (توليد)

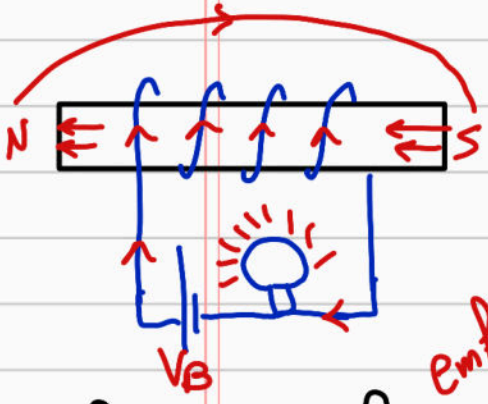
$\phi_m$ — inc. / dec.

emf → Electromotive force

| I ✓ | I = 0 |
|---|---|
| Closed Circuit | open Circuit |
| force ✓ | force = Zero |

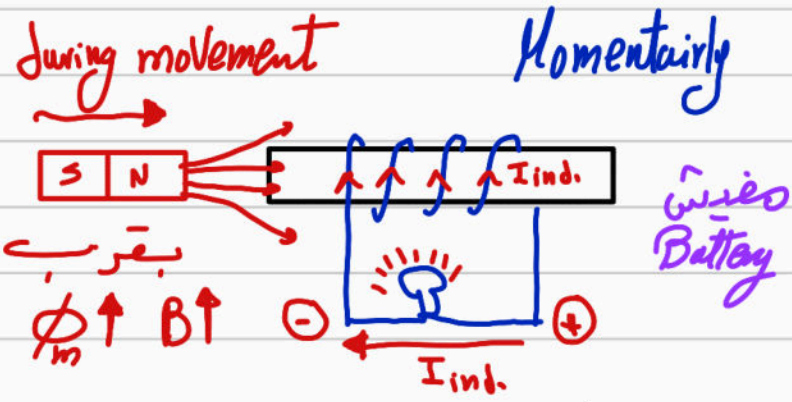
**No Variation** $\phi_m$ ⟶ **No Induction** emf = Zero, I = 0, f = 0

## changable / Varying Magnetic field With respect To Time




$emf_{av} = I R_{ind.}$

during movement — Momentarily

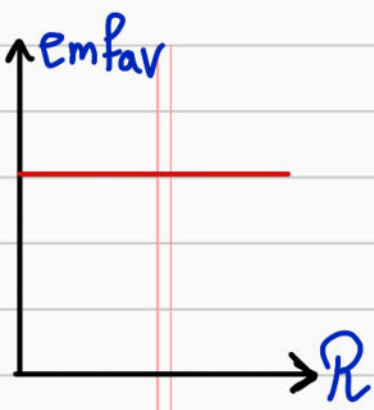


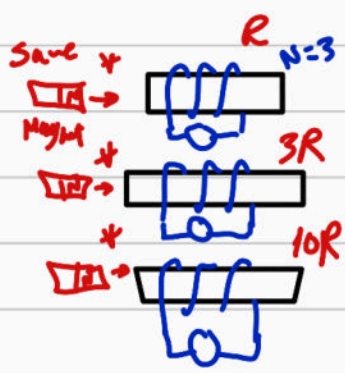

# Faraday's Law: (Magnitude)

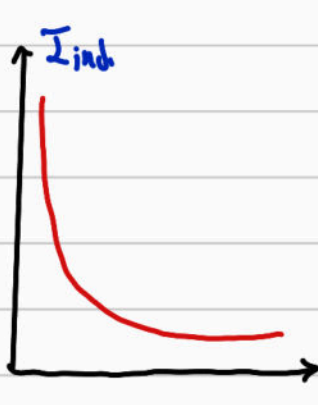
$$emf_{av} = N \frac{\Delta \phi_m}{\Delta t}$$

$\Delta \phi_m \rightarrow \phi_{m2} - \phi_{m1}$ ⇝ Variation ⇝ change

The average induced emf is neumerically equal the Result of Multiplication of the no. of Turns of the coil & the Rate of change of mag. flux acting on the coil.







$$emf_{av} = N \times \frac{\Delta\phi_m}{\Delta t}$$

"factors"







# Lenz's Law: for Direction

The emf is always induced in the direction To **oppose** change Producing it.

$$emf_{av} = -N \times \frac{\Delta\phi_m}{\Delta t} \rightarrow \Delta\phi_m = \phi_{m_2} - \phi_{m_1}$$

$\phi_{m_2} > \phi_{m_1}$ $\boxed{\phi_m \uparrow}$

يبعد ب Magnet الـ

$\Delta\phi_m = +ve$

$emf = -ve$

Repelsion

BackWard induction

$\phi_{m_2} < \phi_{m_1}$ $\boxed{\phi_m \downarrow}$

يجذب Magnet الـ

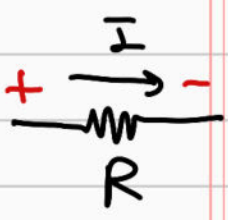
$\Delta\phi_m = -ve$

$emf = +ve$

attraction

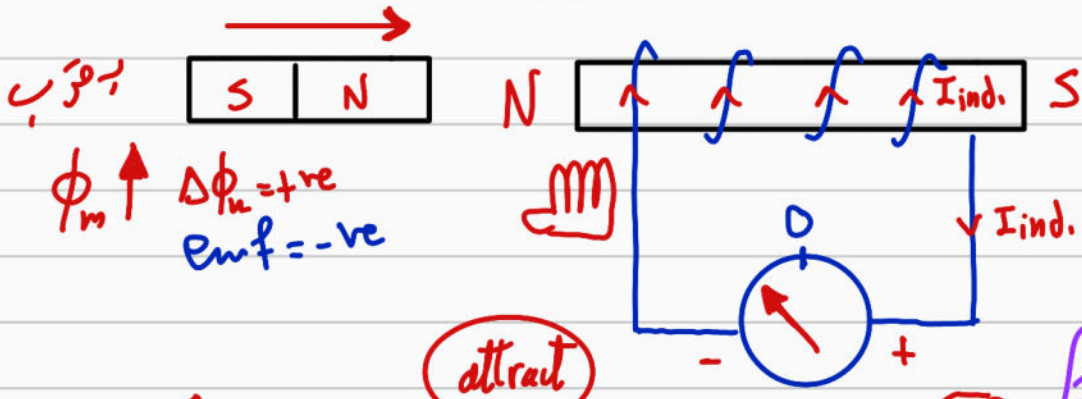
forWard induction

Examples:




Repel

BackWard

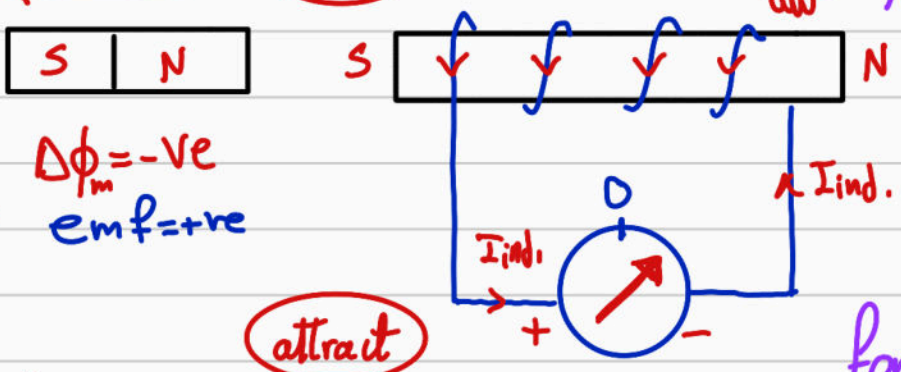



$\phi_m \uparrow$ $\Delta\phi_m$ = +ve

emf = −ve

attract

forWard

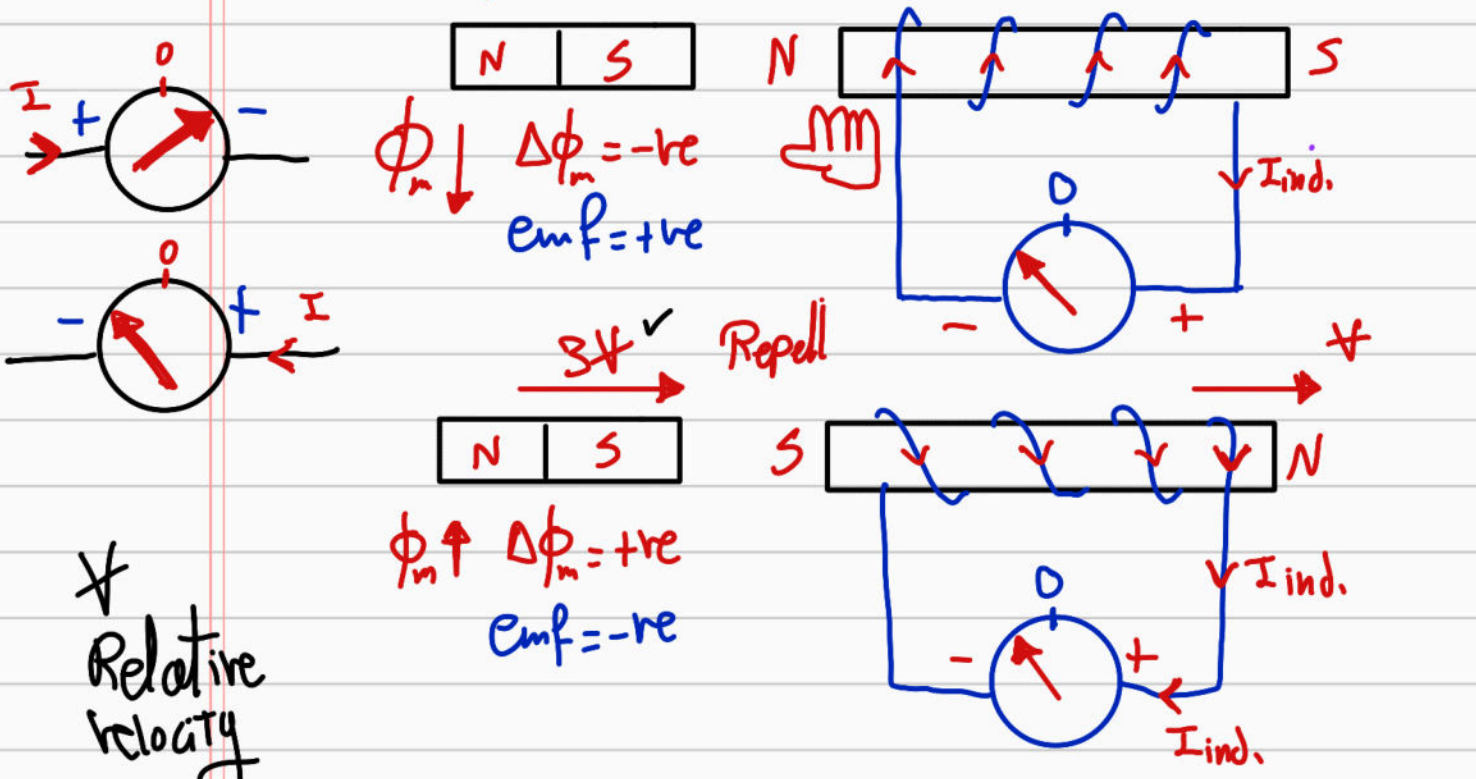



$\phi_m \downarrow$ $\Delta\phi_m$ = −ve

emf = +ve

attract

forWard




$\phi_m \downarrow$ $\Delta\phi_m$ = −ve

emf = +ve

Repell

$\phi_m \uparrow$ $\Delta\phi_m$ = +ve

emf = −ve

Relative velocity

- Same direction: $V_R = V_1 \ominus V_2$
- opposite direction: $V_R = V_1 \oplus V_2$

# Recap:

**Faraday's Law: (Magnitude)** $emf_{av} = N \times \frac{\Delta \phi_m}{\Delta t} = N \times \frac{[\phi_{m_2} - \phi_{m_1}]}{\Delta t}$

$Q \times R = N \times \Delta \phi_m$

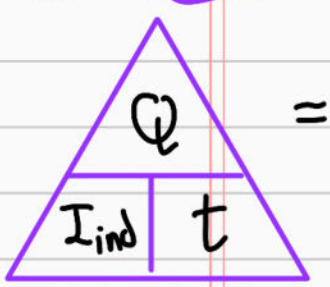
$emf_{av} = I_{ind.} \times R$

$I_{ind.} R = N \times \frac{\Delta B_{ex} A}{\Delta t} = N \frac{[B_2 A_2 - B_1 A_1]}{[t_2 - t_1]}$

$\frac{Q}{t} R = NB \times \frac{\Delta A}{\Delta t} = NB \frac{[A_2 - A_1]}{\Delta t}$




$= NA \frac{\Delta B_{ex}}{\Delta t} = NA \frac{[B_2 - B_1]}{\Delta t}$




$= NBA \frac{\Delta \theta}{\Delta t} = NBA \frac{[\sin\theta_2 - \sin\theta_1]}{\Delta t} = NBA \frac{[\cos\theta_2 - \cos\theta_1]}{\Delta t}$



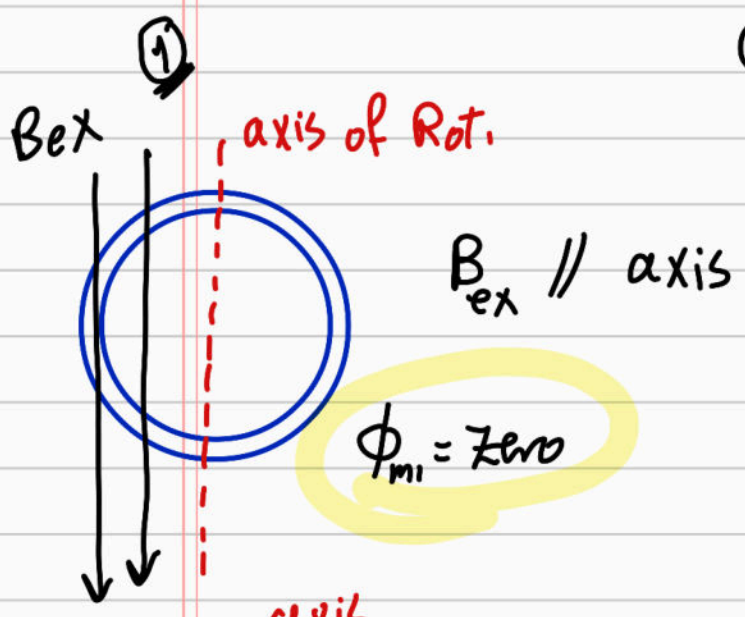

**Lenz's Law: (Direction)** $emf_{av} = -N \times \frac{\Delta \phi_m}{\Delta t}$

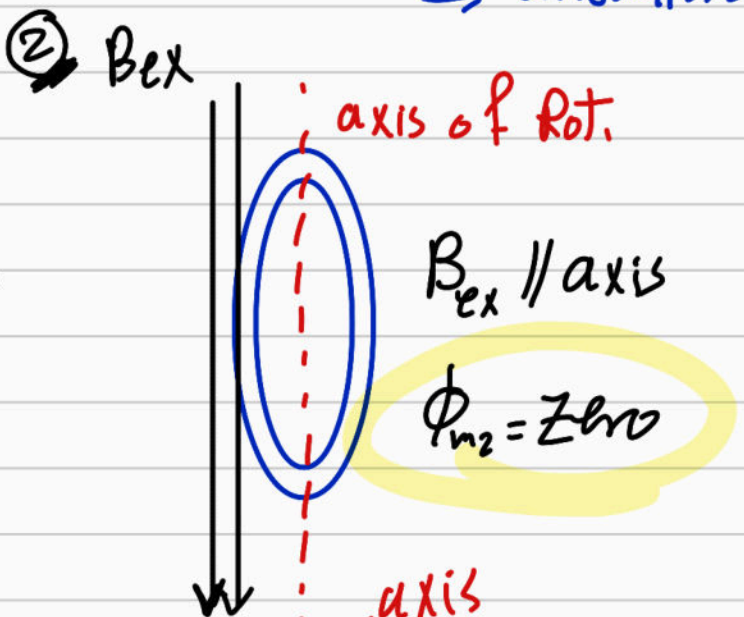
↳ emf is always induced in the direction To oppose ✓ change producing it.

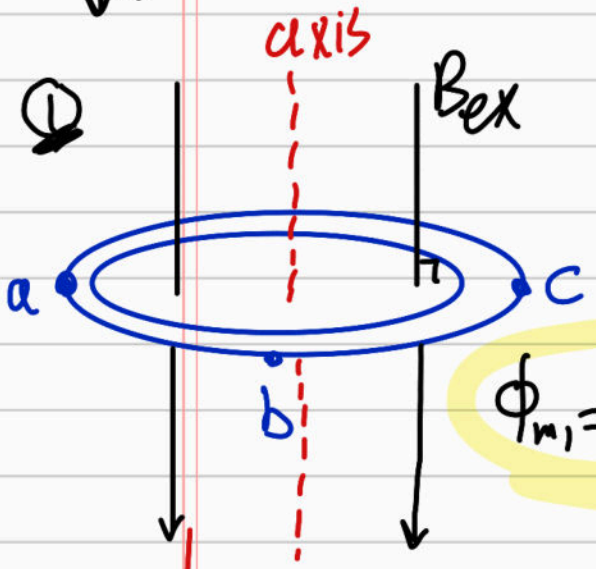
direction

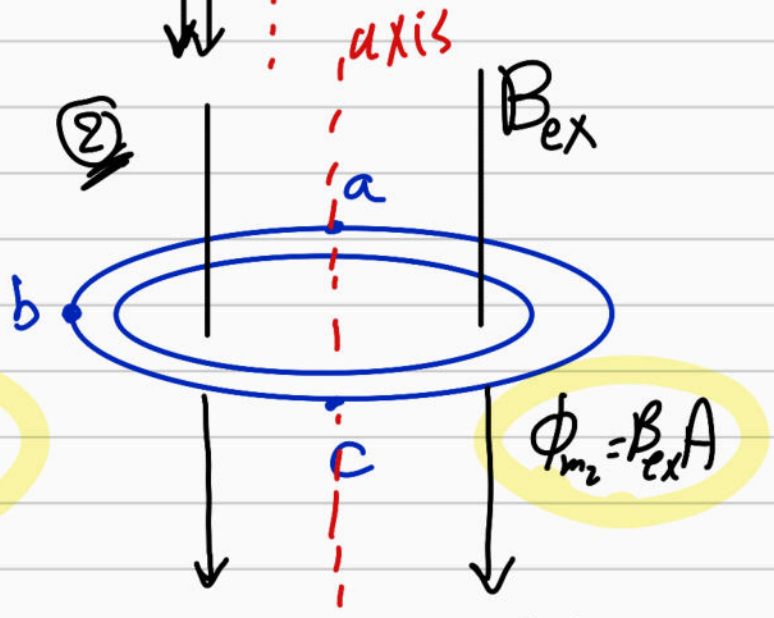
| forWard | Backward |
|---|---|
| $\phi_m \downarrow$ Move away | $\phi_m \uparrow$ Move Closer |
| $B_{ex} \downarrow$ | $B_{ex} \uparrow$ |
| $V_B \downarrow \propto I \downarrow$ open switch | $I \uparrow$ close switch |
| $A \downarrow$ | $A \uparrow$ |
| $R \uparrow$ Rehostate inc. | $R \downarrow$ Rehostate dec. |
| $\Delta \phi_m = -ve \rightarrow emf = +ve$ | $\Delta \phi_m = +ve \rightarrow emf = -ve$ |

Note: $B_{ex} \perp$ axis of Rotation $\Rightarrow$ Variation $\Rightarrow$ Induction

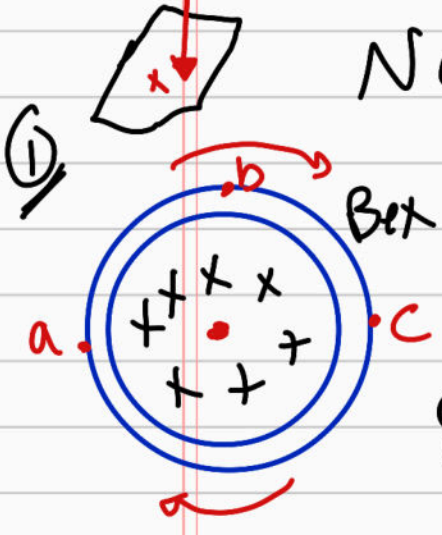


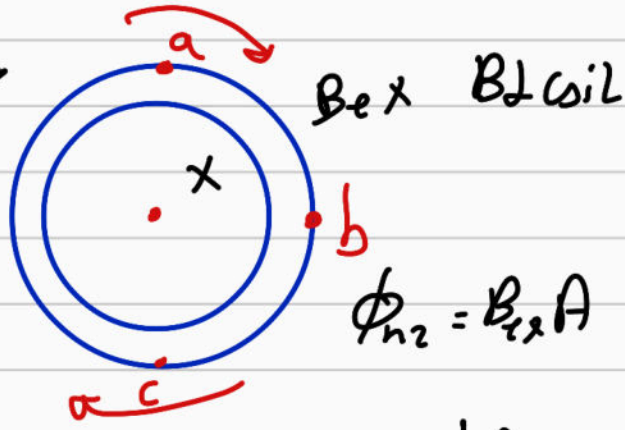



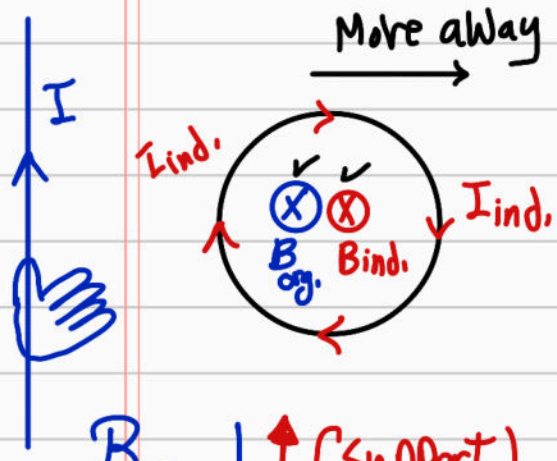

No Variation ≡ No Induction



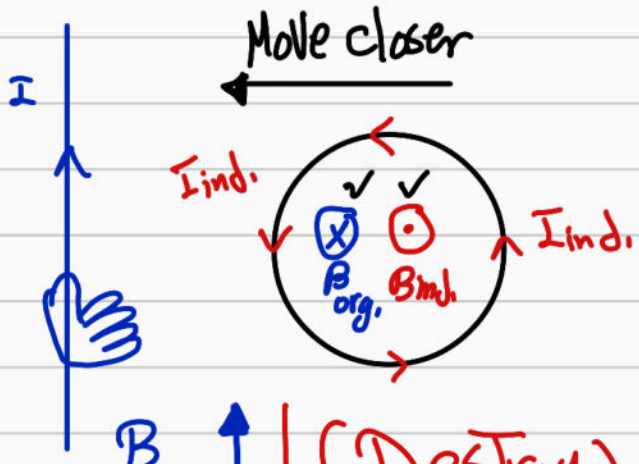



## forWard

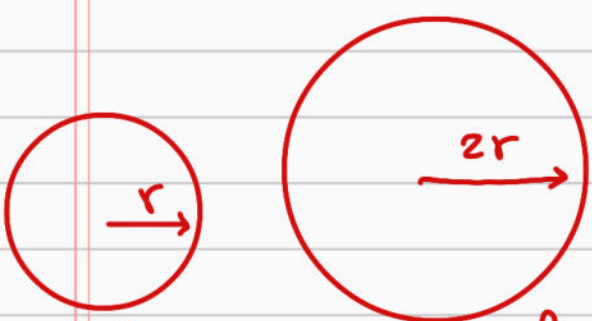



$B_{org.}$ ↓ ↑ (support)

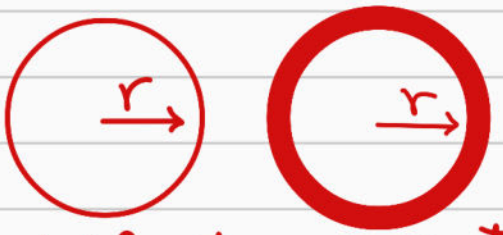
## BackWard




$B_{org.}$ ↑ ↓ (DesTroy)

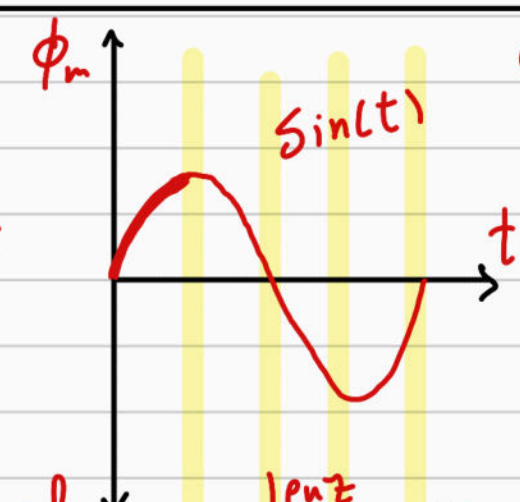


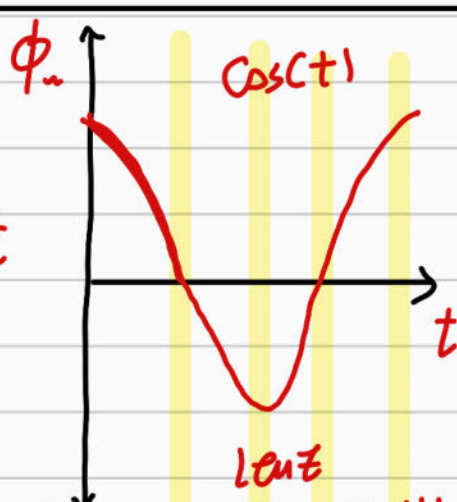

uniform cross section of Wire




different cross section of Wire



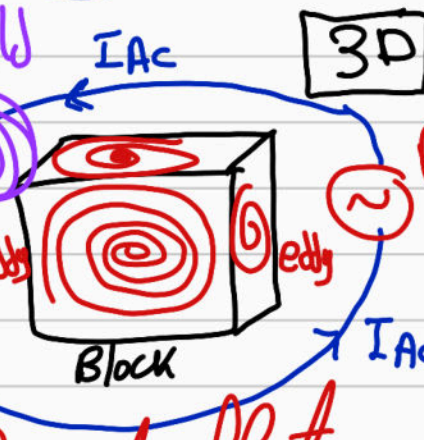



# 3 Eddy Current:

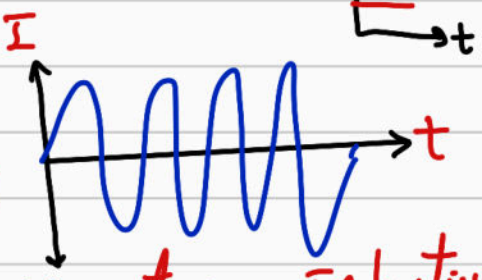
1. Metal Block ✓
2. Metal sheet ✓
3. Metal core ✓

| | |
|---|---|
| | x |
| | x |
| | x |

No eddy Current




AC Source ✓✓
↳ Alternating Current



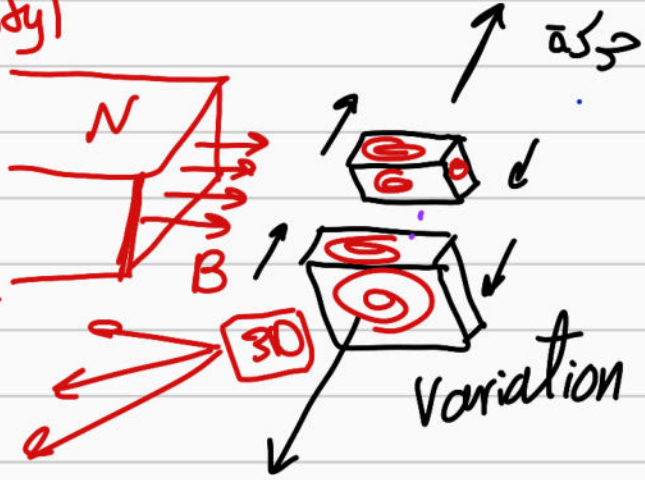

Variation → Induction ✓✓

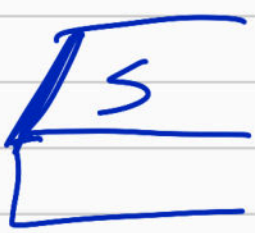
high Thermal effect ⇒ Induction furnace

Ac → ΔB → Ieddy ↑ → Temp. ↑

Electric energy → Magnetic energy → Electric energy → thermal energy

Temp. ↑ ∝ Ieddy ↑




Variation ✓
$\Phi_m$ / I / B / A
Induction (eddy Current)

## # Factors affecting eddy Current!

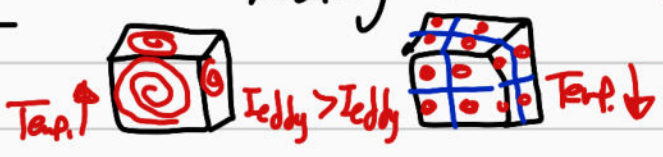
$R = \frac{\rho L}{A}$

↓ $I = \frac{V}{R}$ → ↑ $\frac{\rho l}{A}$

$I \propto \frac{1}{\rho} \propto \sigma$

1. $I_{eddy} \propto$ Freqeuny Ac Source
2. $I_{eddy} \propto \sigma$ conductivity $\propto \frac{1}{\rho}$ Restivity $\propto \frac{1}{R}$
3. $I_{eddy} \propto \frac{1}{\text{Thin/Insulated sheets}}$

A ↓ → R ↑ → I ↓ → Temp. ↓

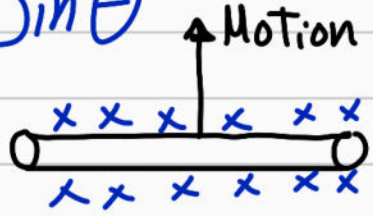


# 4 emf Straight Wire;

## 1 Magnitude;

$$emf_{Wire} = -B_{ex} l V \sin\theta$$

(− → Lenz direction)

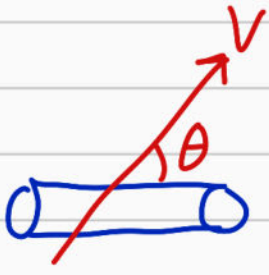



emf → Electromotive force induced in straight wire [Volt]

$B_{ex}$ → external Magnetic flux density [Tesla]

$l$ → length of wire subjected To magnetic field. [meter]

$V$ → velocity of Wire [m/sec]




$\theta$ → angle between Wire & $B_{ex}$
Wire & velocity
$B_{ex}$ & velocity

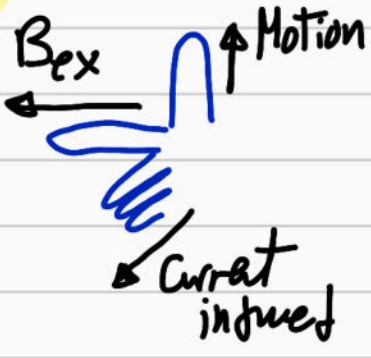
$$emf_{wire} = B l V \sin\theta = I R$$

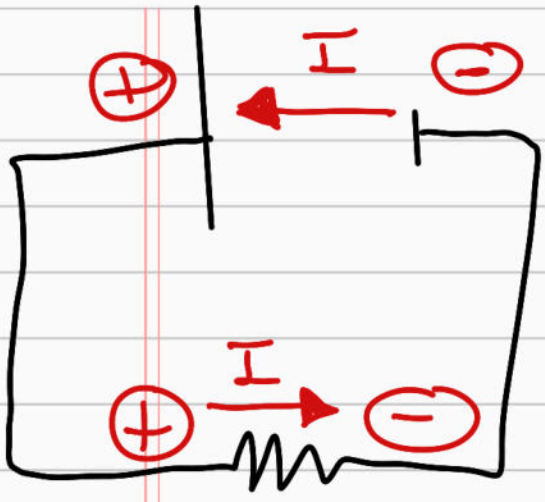
## 2 Direction; Fleming Right hand Rule

Thumb → Motion / velocity (Force)

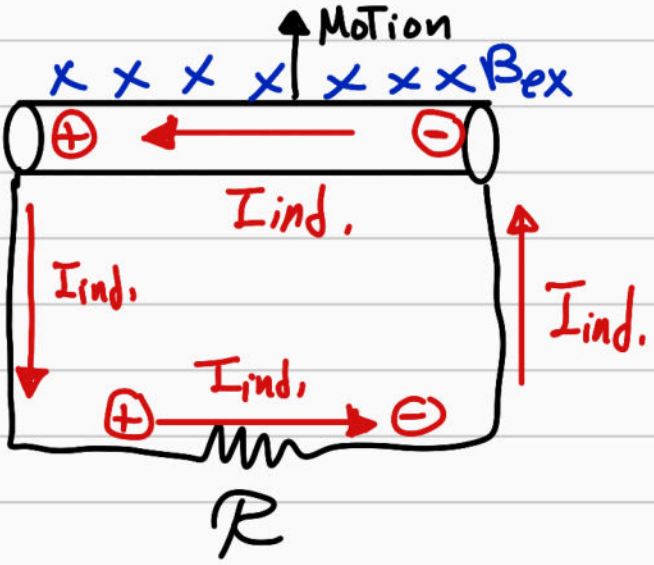
Fore[pointer] → $B_{ex}$

Rest of fingers → Curret induced

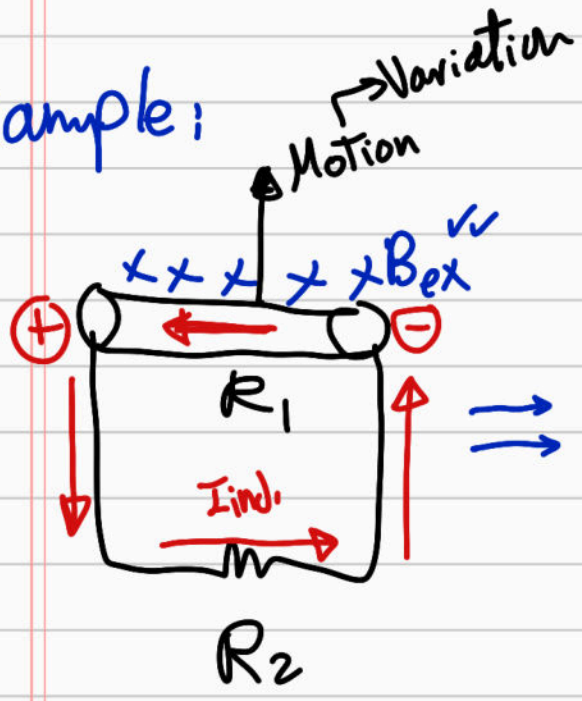


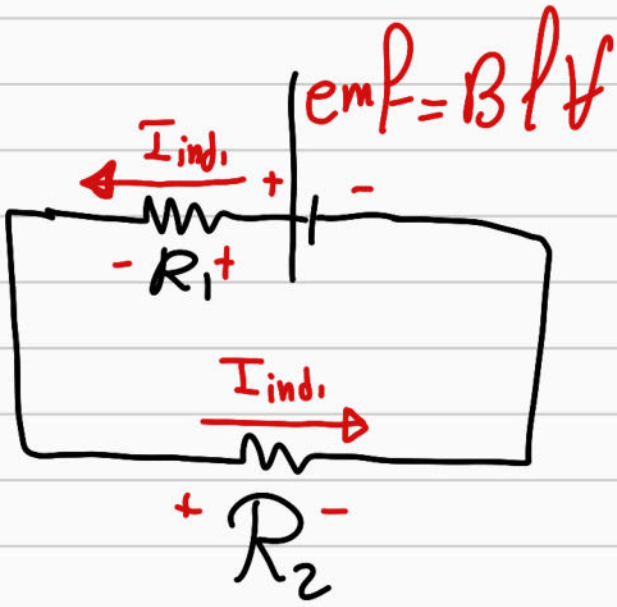



The wire acts like a Battery




#Example:



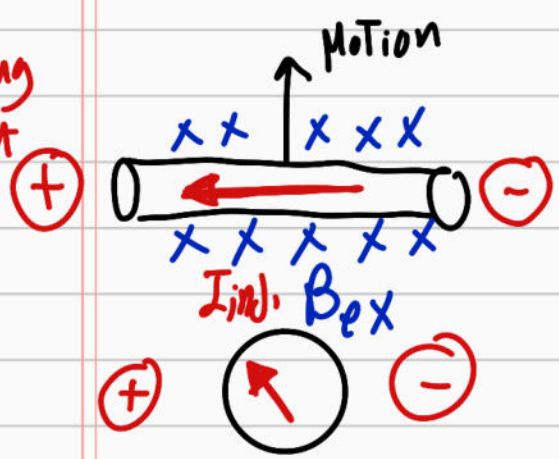




#Force

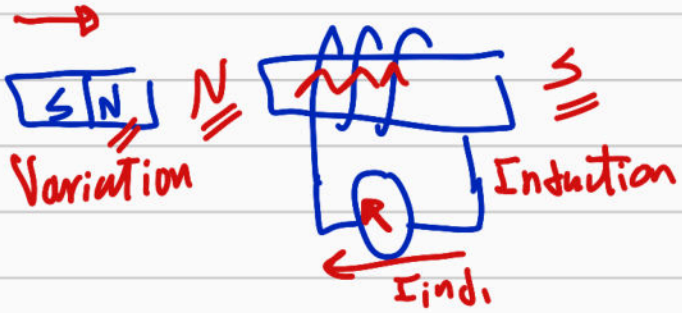
$$F = |IB_{ex}| = \frac{B_{ex}^2 L^2 v}{R} = \frac{[B_{ex} L v]^2}{Rv} = \frac{emf^2}{Rv}$$

$$F = \frac{P_w}{v} = \frac{I^2 R}{v}$$

Fleming Right




Mutual



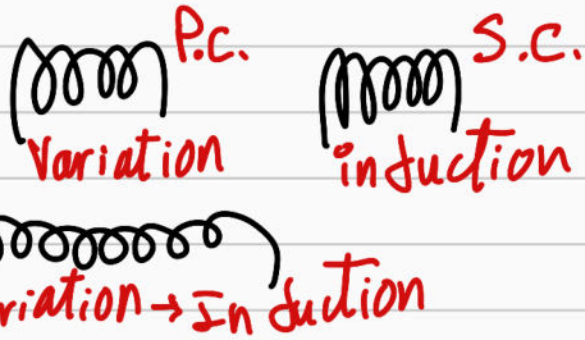
Chapter ③ Lesson ②

↳ Coil & Solenoid

# Mutual & self Induction

Mutual → Two different coils.

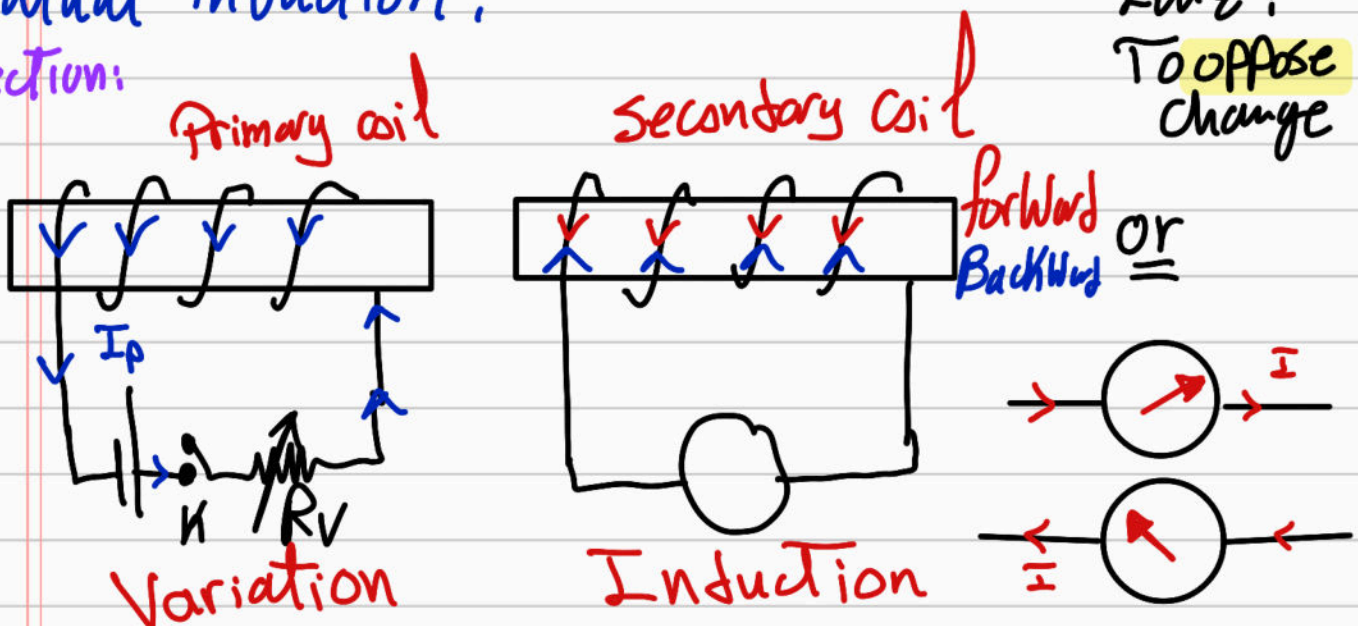
Self → Same coil.




## 1 Mutual induction:

Lenz: To oppose change

*Direction:

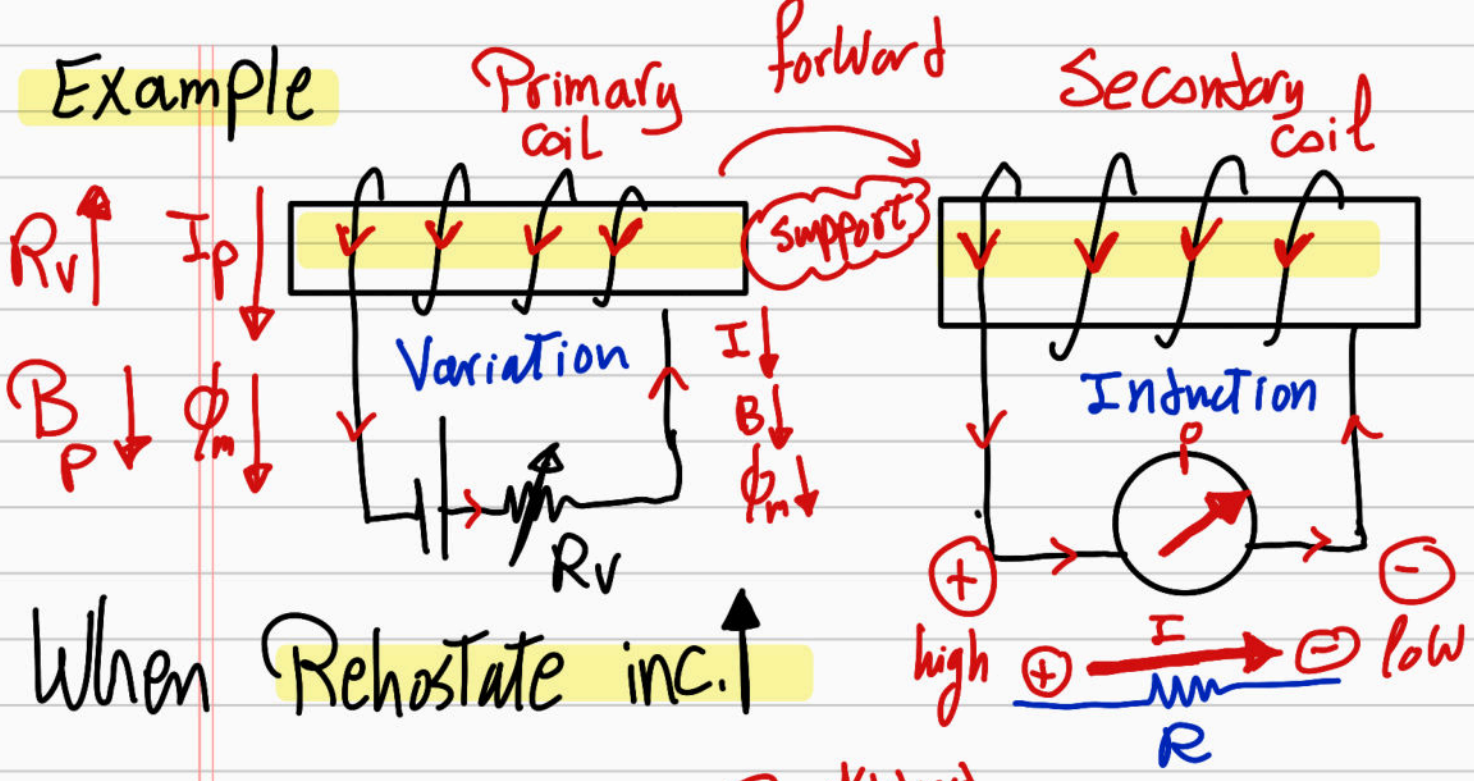



| forWard [support] | BackWard [destroy] |
|---|---|
| $I_{ind.}$ s.c. same as $I_{p.c.}$ | $I_{ind.}$ s.c. opposite to $I_{p.c}$ |
| $\phi_m \downarrow$ فلوى<br>$B \downarrow$ نجاح<br>$I \downarrow$ موجة<br>$R \uparrow$ مقاومة | $\phi_m \uparrow$ فلوى<br>$B \uparrow$ نجاح<br>$I \uparrow$ موجة<br>$R \downarrow$ مقاومة |

دينا معاك

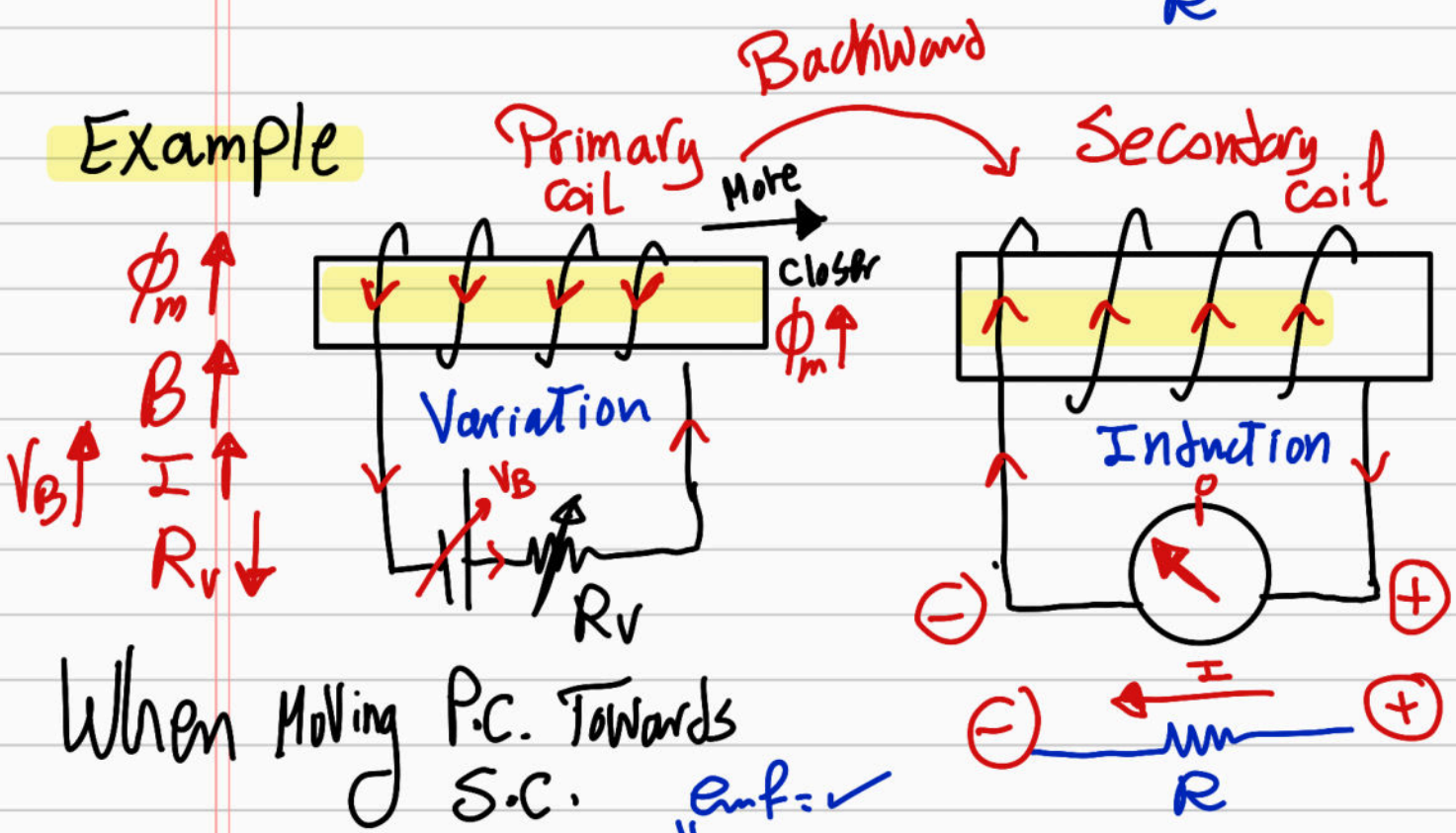
حسد حقد

## Example




When Rehostate inc.↑

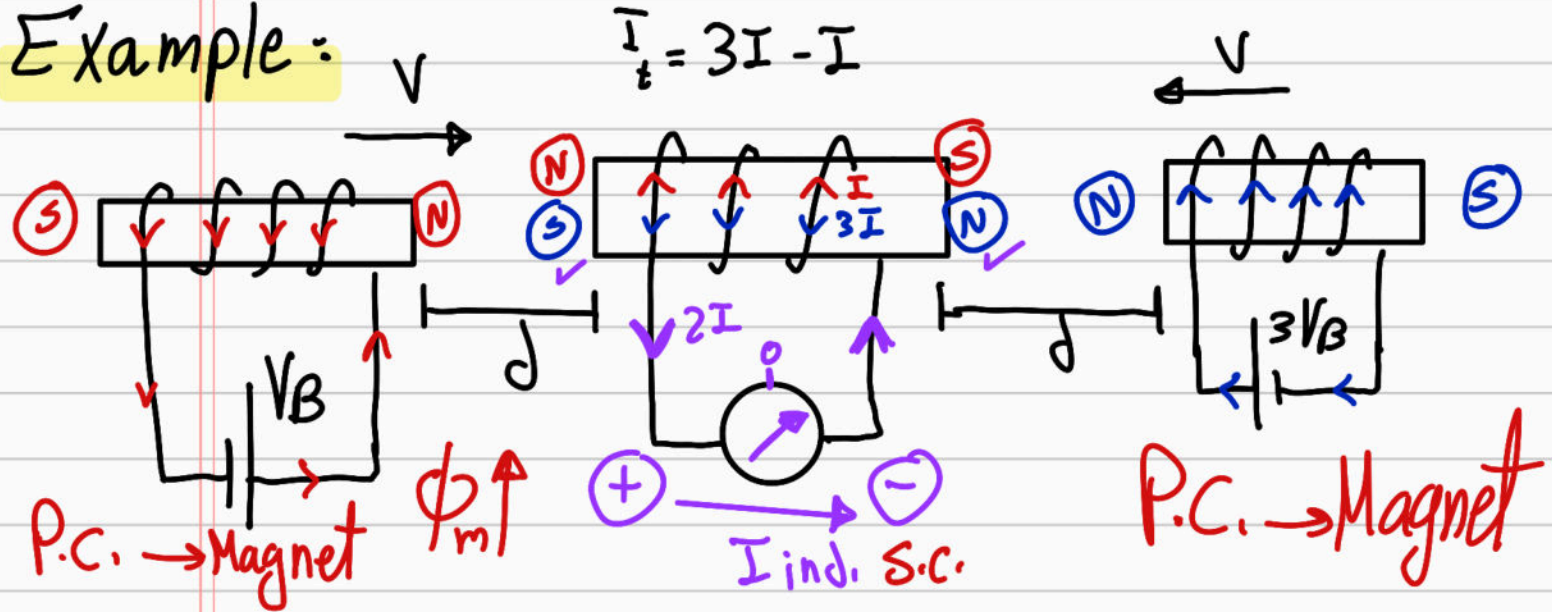
## Example




When Moving P.C. Towards S.C.

emf: ✓
$V_B = 0$

## Example:



* Magnitude: $\Phi_m = B_{ex} A$ → $B_p A_s$

P.C. → Variation
S.C. → Induction

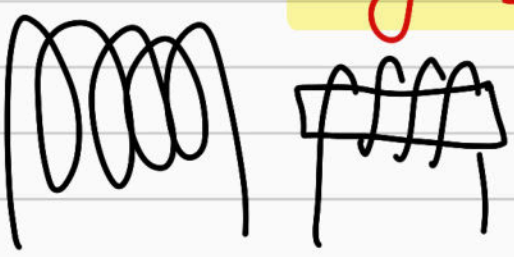
1. $emf_{s_{av}} = -N_s \frac{\Delta \Phi_{m_s}}{\Delta t}$ (− : Lenz)

2. $emf_{s_{av}} = I_{ind_s} R_s$

3. $emf_{s_{av}} = -M \frac{\Delta I_{Battery_p}}{\Delta t}$ (− : Lenz)

$M$ → Mutual Induction coff.
Mutual Inductance → Henry [H]

4. $N_s \Delta \phi_{m_s} = M \Delta I_p$



5. $M = \frac{\mu_p A_s N_p N_s}{l_{s_p}}$ or $\frac{\mu_p A_s N_p N_s}{2 r_p}$

($\mu$ : Premability; $l_{s_p}$ ⇒ sol.; $r_p$ ⇒ coil)




**Proof**

$M \Delta I_p = N_s \Delta \phi_s \rightarrow B_p A_s$

$M \Delta I_p = N_s A_s \Delta B_p$ → coil / sol.

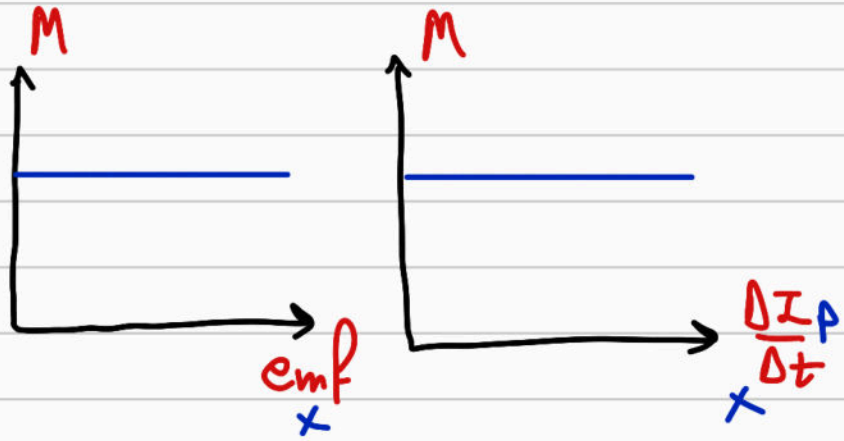
$M \cancel{\Delta I_p} = N_s A_s \times \frac{\mu_p N_p \cancel{\Delta I_p}}{l_{sp}}$

$$M = \frac{\mu_p A_s N_p N_s}{l_{sp}}$$

# *factors [M] :

false factors

$$M = \frac{emf}{\frac{\Delta I_p}{\Delta t}}$$

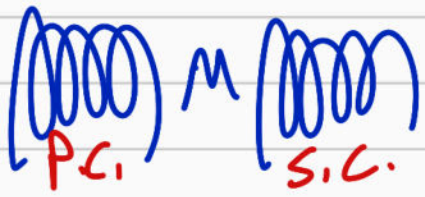



factors:

$$M = \frac{\mu_p A_s N_p N_s}{l_{sp} \text{ or } 2r_p}$$

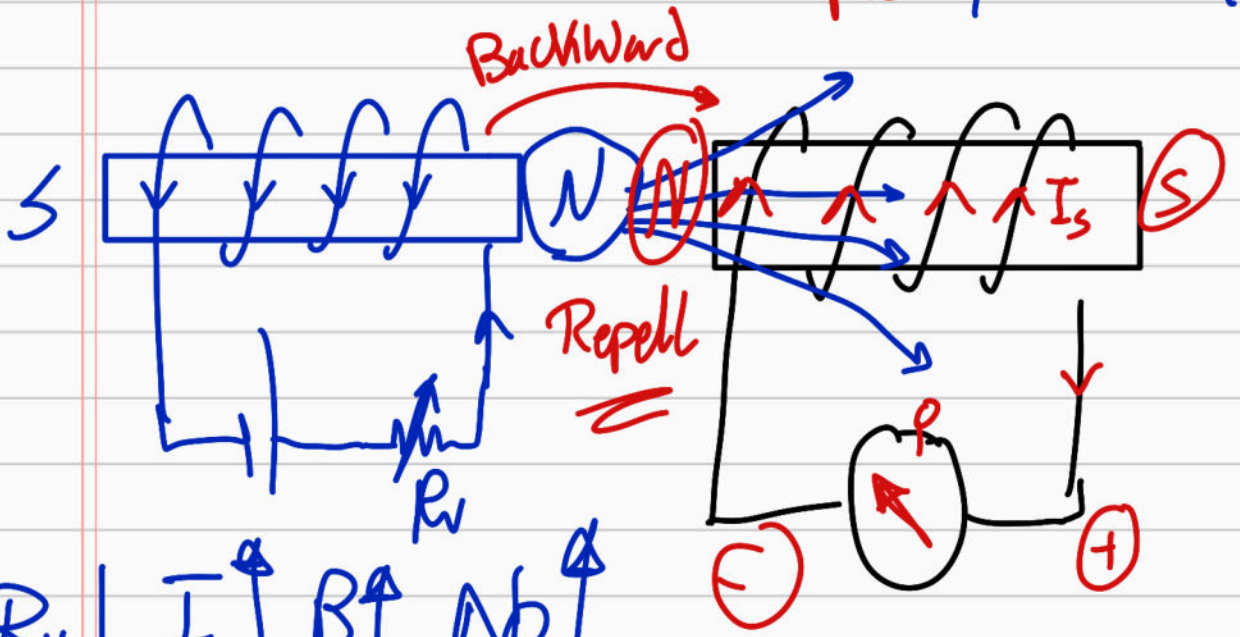
$M \downarrow \quad M \propto \frac{1}{\text{distance}} \uparrow$

1. Area of secondary coil
2. distance between coils
3. No. of Turns $\begin{cases} \text{P.C.} \\ \text{S.C.} \end{cases}$





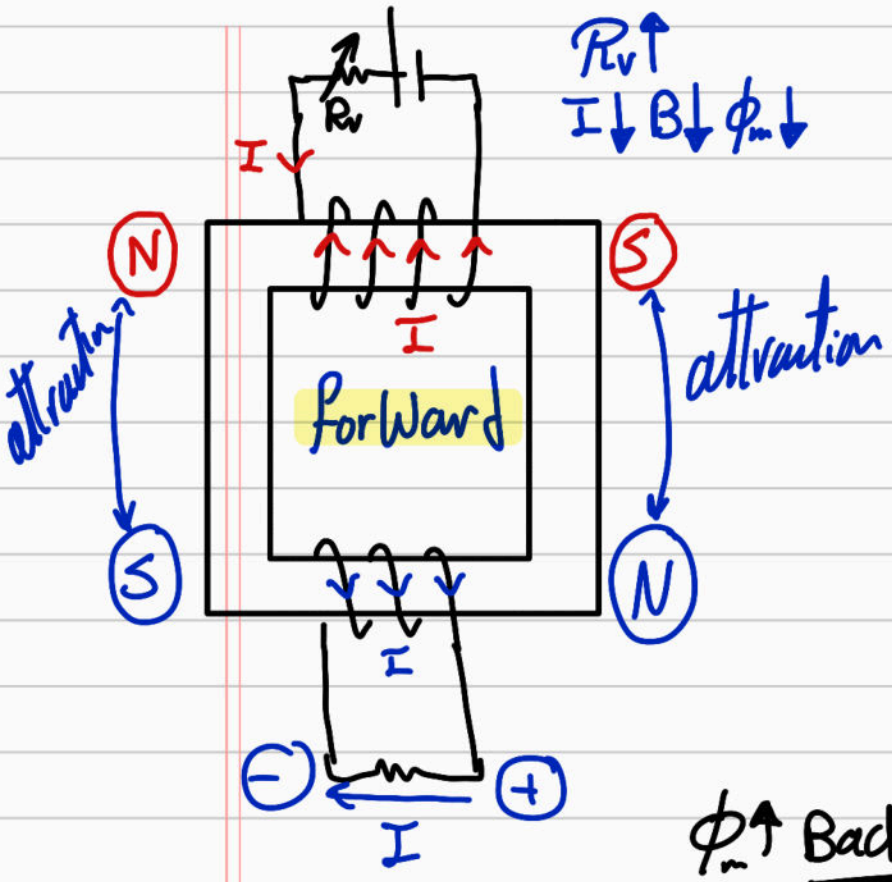


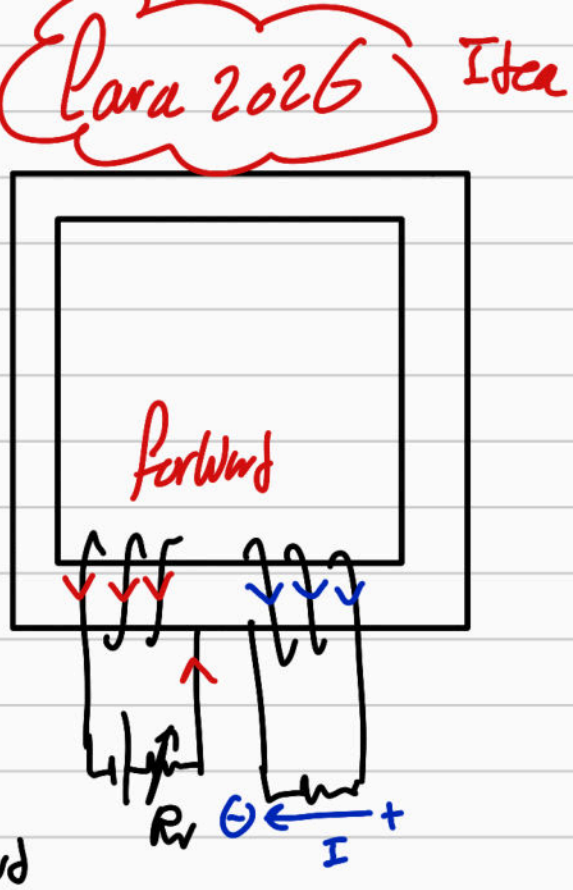



$R_v \downarrow \quad I_p \uparrow \quad B_p \uparrow \quad \Delta p_m \uparrow$

Lara 2026 Idea

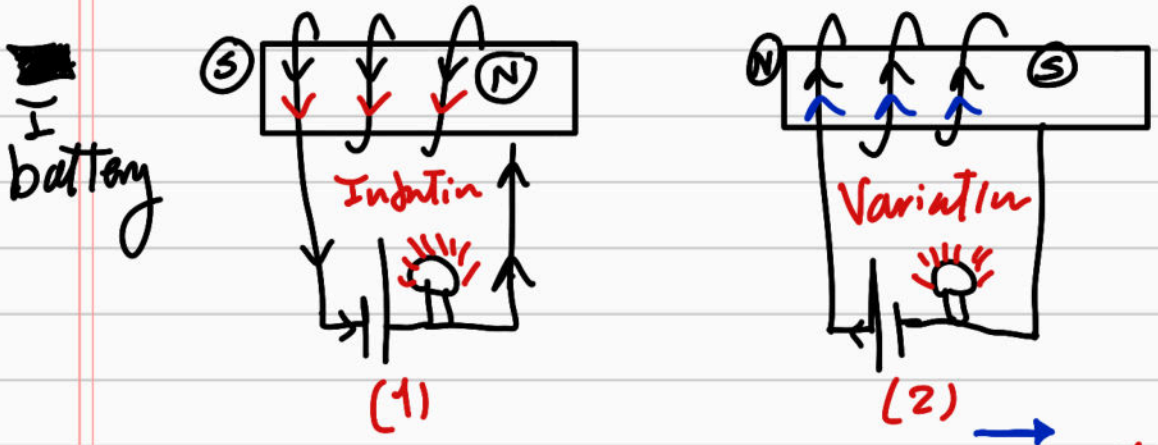
$R_v \uparrow$
$I \downarrow B \downarrow \phi_m \downarrow$

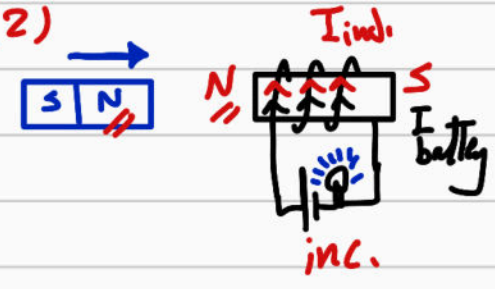



$\phi_m \uparrow$ Backward

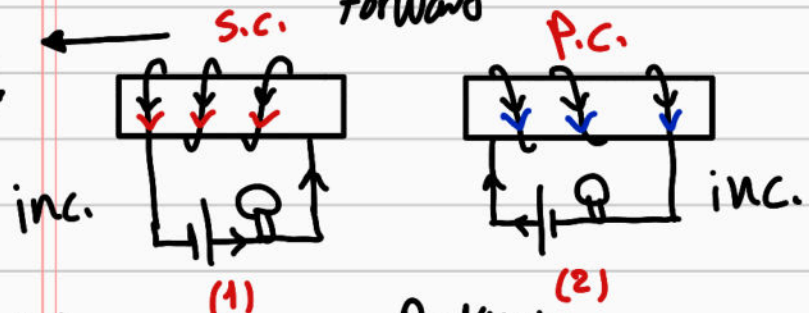
S.C. → P.C.



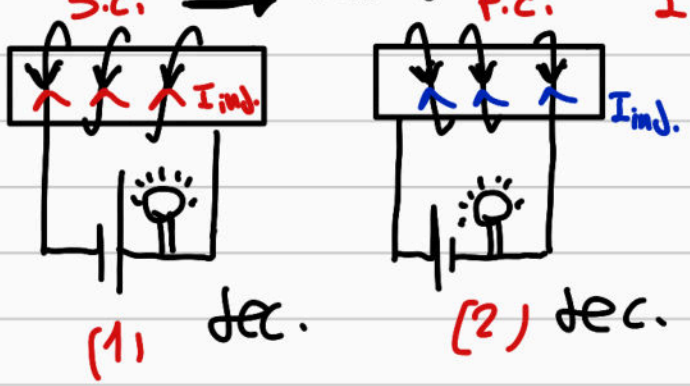




$I_{ind.}$

$\phi_m \downarrow$ ← S.C. ForWard P.C.




$\phi_m \uparrow$ S.C. → Backward P.C.




$I = \frac{V_B}{R}$ $I_{ind.} = \frac{\mathcal{E}_{ind}}{R}$

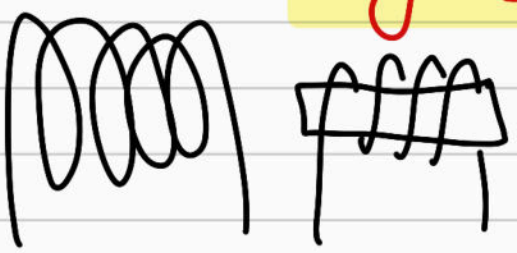
# 2 Self Induction:

## 1 Rules:

1. $emf_{av} = -N \frac{\Delta \phi_m}{\Delta t}$ (− : Lenz)
2. $emf_{av} = I_{ind.} R$
3. $emf_{av} = -L \frac{\Delta I}{\Delta t}$ (− : Lenz)
4. $N \Delta \phi_m = L \Delta I$
5. $L = \frac{\mu A N^2}{l_s}$ or $\frac{\mu A N^2}{2r}$ ($\mu$: Premability)

$L$ → Self Induction coff. / Self Inductance

Henry [H]



$l_s$ ⇒ sol. (solenoid)

$r$ ⇒ coil

$emf = N \frac{\Delta \phi_L}{\Delta t}$

$V = \frac{Wb}{s}$

$\phi_m = BA$

$Wb = T \cdot m^2$

$B = \frac{\mu I}{2\pi d} \cdot \frac{Wb}{A \cdot m}$

$T = \frac{\mu \cdot A}{m} \rightarrow \mu = \frac{T \cdot m \times m}{A \times m}$

## # Equivalent Units:

$emf = L \frac{\Delta I}{\Delta t}$

$V = H \cdot \frac{A}{s}$

$H = \frac{V \cdot s}{A}$

$H = \frac{V \cdot s}{A} = \Omega \cdot s$

$= \frac{Wb}{A} = \frac{T \cdot m^2}{A}$

(A) → $\frac{V}{\Omega}$, $\frac{W}{V}$, $\frac{C}{s}$

$H = \ldots\ldots$ $\quad W = VIt$

(a) $\Omega/s \rightarrow \Omega.s$

(b) $J/A \rightarrow \frac{V.A.s}{A} = V.s = Wb$

(c) $\frac{T.m^2}{A.s} \rightarrow \frac{Wb}{A.s} = \frac{H}{s} \neq H$

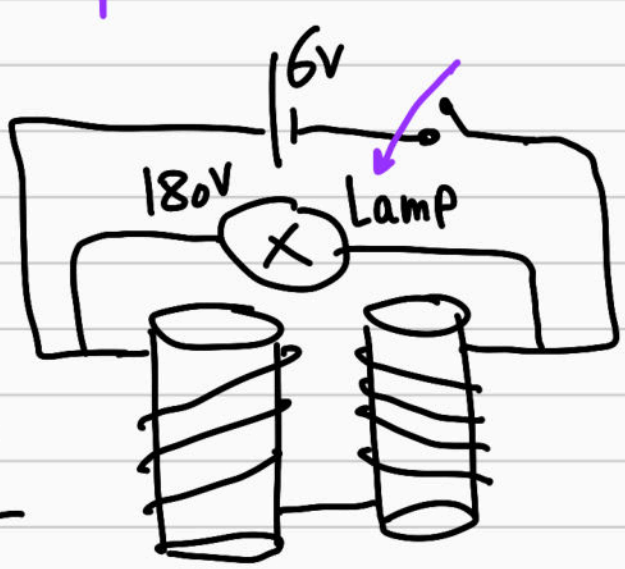
(d) $\frac{N.m}{A^2} \rightarrow \frac{m.A.T.m}{A^2} = \frac{T.m^2}{A} = \frac{Wb}{A} = H$

## 2 Self Induction Exprment:

Switch on: Key closed

$I \uparrow B \uparrow \phi_m \uparrow \rightarrow$ BackWard

The lamp doesn't Glow.



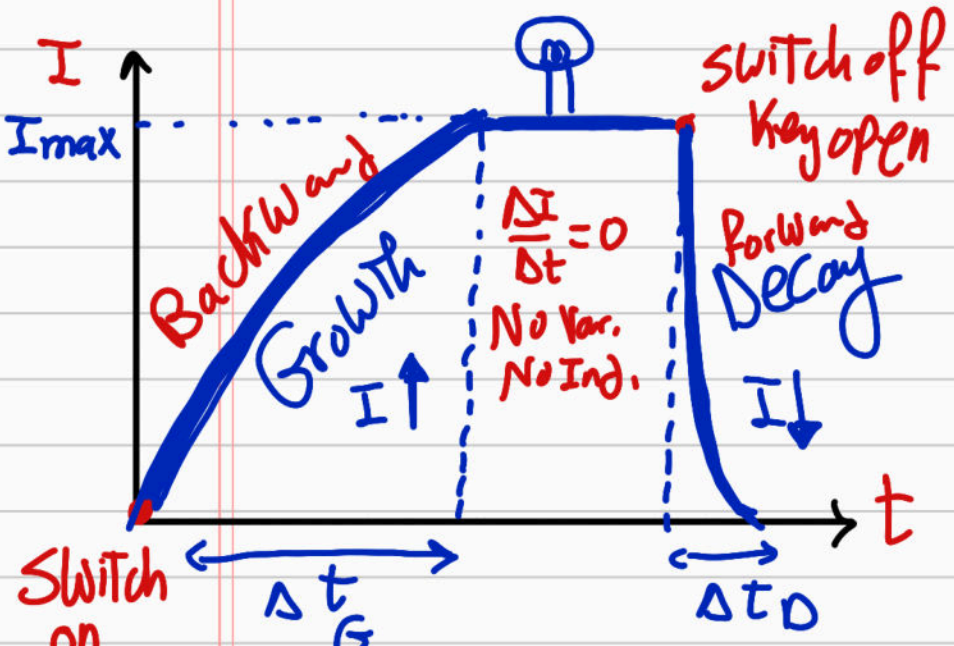

Switch off: Key open

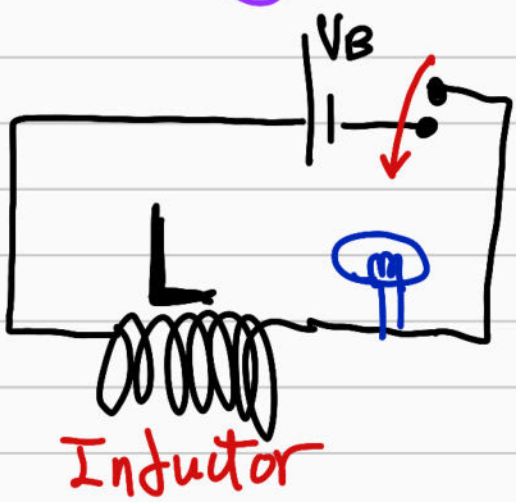
$I \downarrow B \downarrow \phi_m \downarrow \rightarrow$ forWard

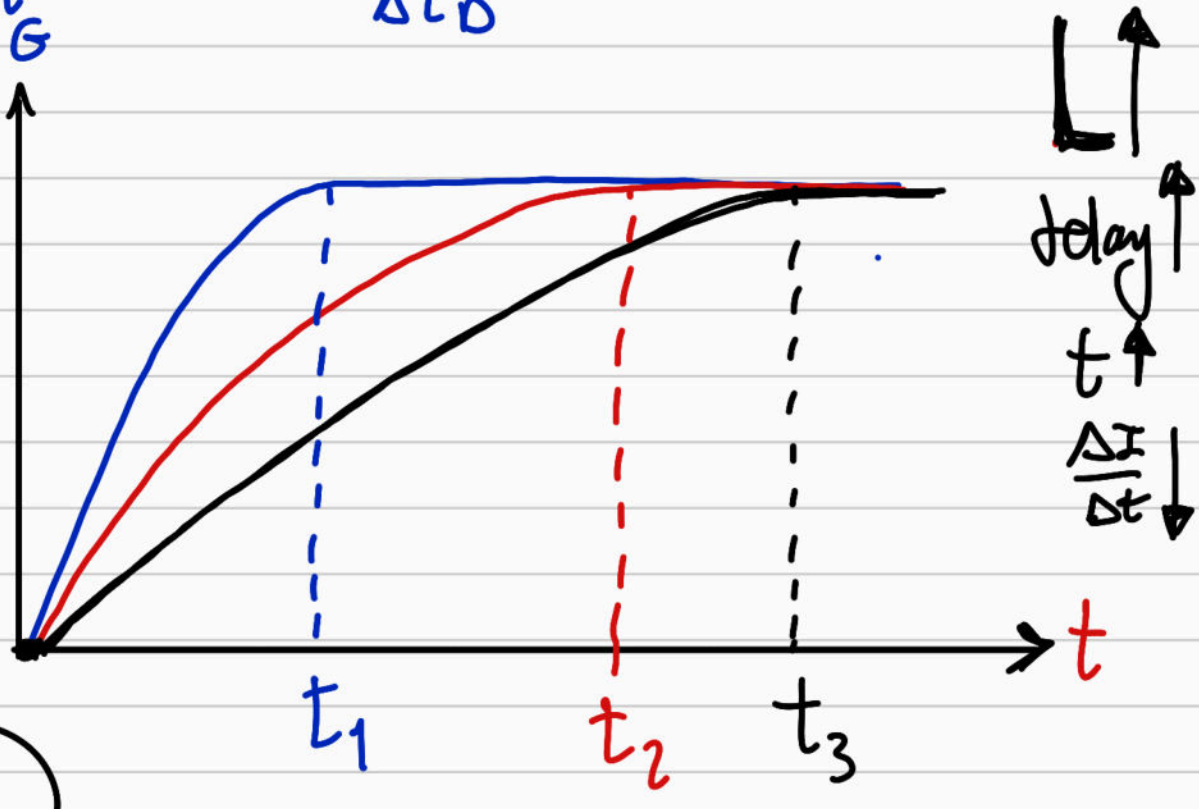
The lamp Glows momentarily then Vanishes

$\Delta t_G \uparrow\uparrow$ $\quad emf_{Back} \downarrow\downarrow \ll\ll emf_{forWard} \uparrow\uparrow$ $\quad \Delta t_o \downarrow\downarrow$

# 3 Rate of Growth & Decay




I ↑

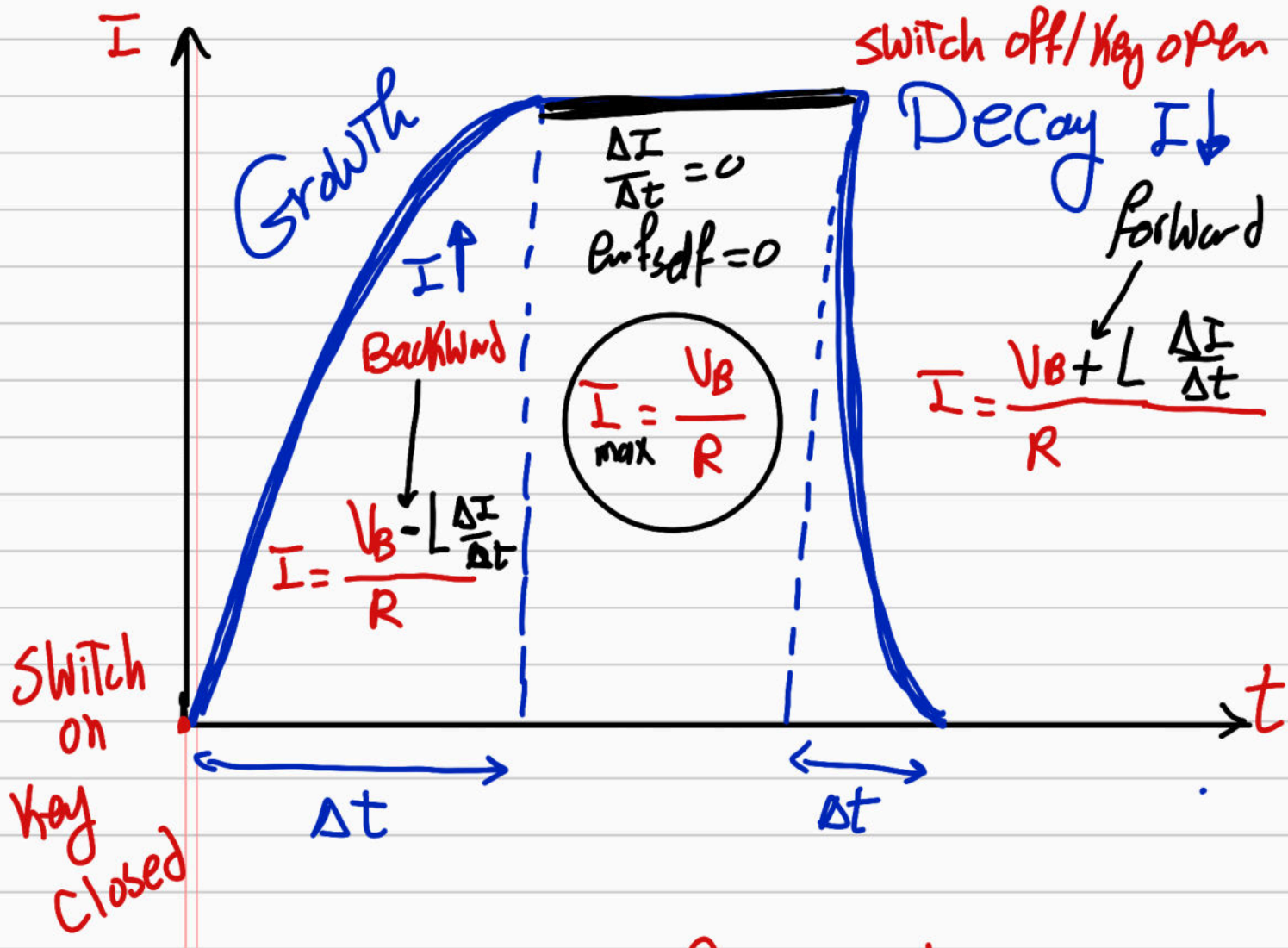
Backward
↳ delay

$t_1 < t_2 < t_3$

$\frac{\Delta I}{\Delta t}_1 > \frac{\Delta I}{\Delta t}_2 > \frac{\Delta I}{\Delta t}_3$

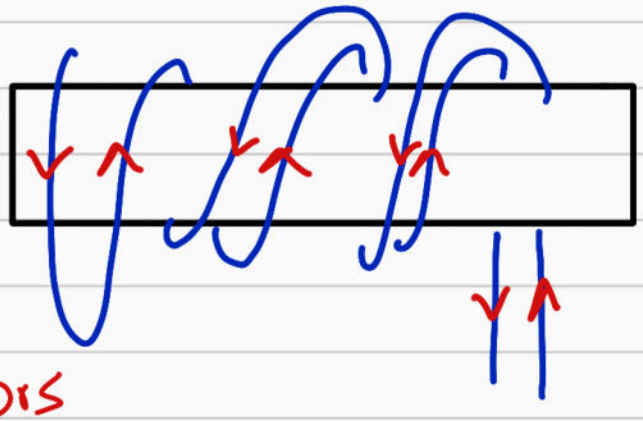
$L_1 < L_2 < L_3$

↙ Self Induction eff.




# #When To Cancel self Induction :

$$L = 0$$



* Calibrated Resistors
* Double Wonded Wires

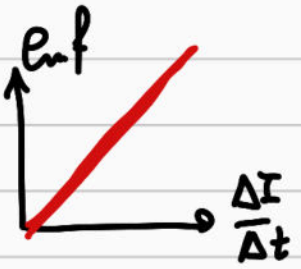
factors

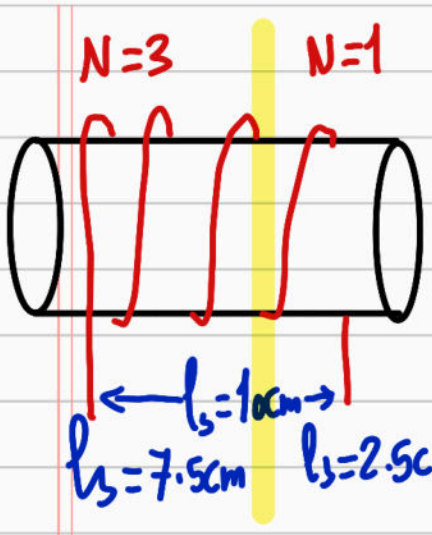
$$L = \frac{\mu A N^2}{l_s}$$




→ constant

$$L = \frac{emf}{\frac{\Delta I}{\Delta t}}$$



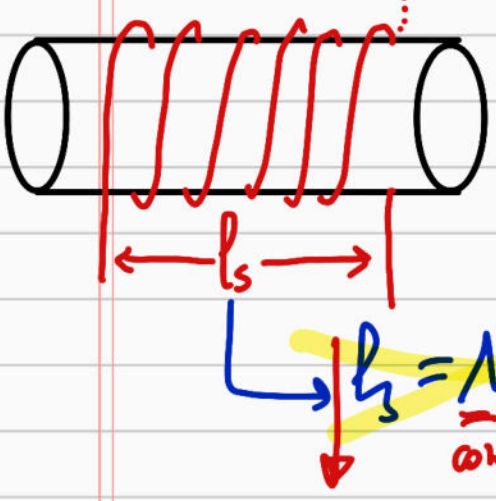



$$L = \frac{\mu A \left[N\left(\frac{3}{4}\right)\right]^2}{l_s\left(\frac{3}{4}\right)} = \frac{\left(\frac{3}{4}\right)^2}{\left(\frac{3}{4}\right)} = \frac{3}{4}$$

$$\frac{3}{4}L = \frac{\mu A N N \frac{3}{4}}{l_s \frac{3}{4}}$$

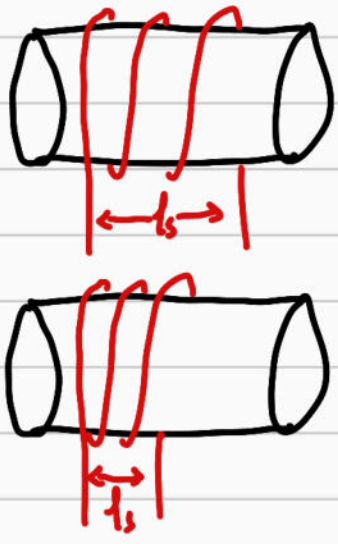
$L_{wire}$

$$N = \frac{L_{wire}}{2\pi r}$$



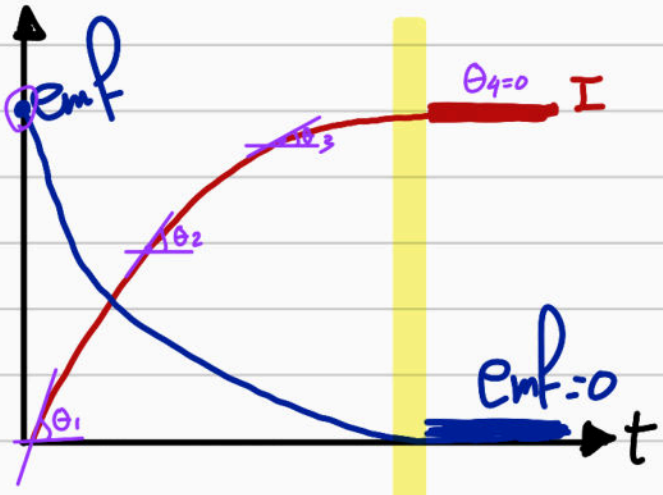

$l_s = N \times 2\pi r$

const. const.










$$Rate = \frac{\Delta I}{\Delta t} = \text{slope}$$

$\theta_1 > \theta_2 > \theta_3 > \theta_4 = 0$

slope 1 > slope 2 > slope 3 > slope 4

$\tan\theta_1 > \tan\theta_2 > \tan\theta_3 > \tan\theta_4$

$\frac{\Delta I}{\Delta t}1 > \frac{\Delta I}{\Delta t}2 > \frac{\Delta I}{\Delta t}3 > \frac{\Delta I}{\Delta t}4 = 0$

$emf_1 > emf_2 > emf_3 > emf_4$

# "Ac Dynamo"

مولد كهربي

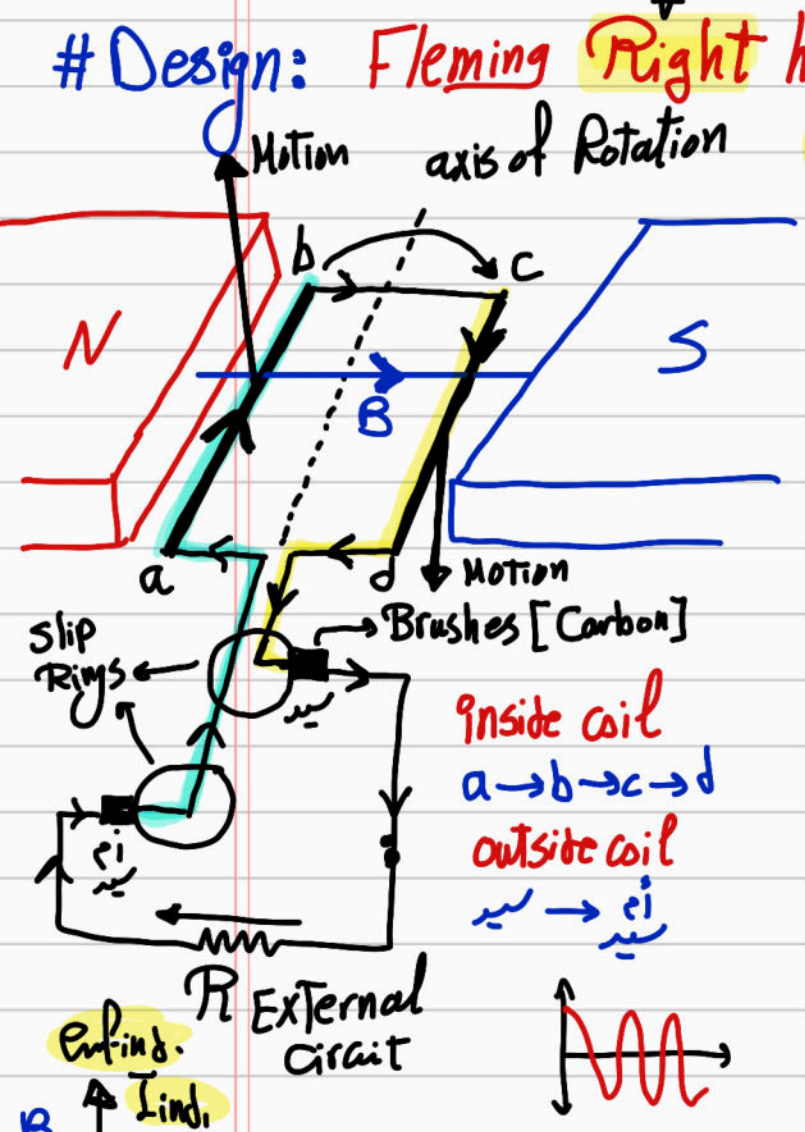
#other names: Genrator - Alternator

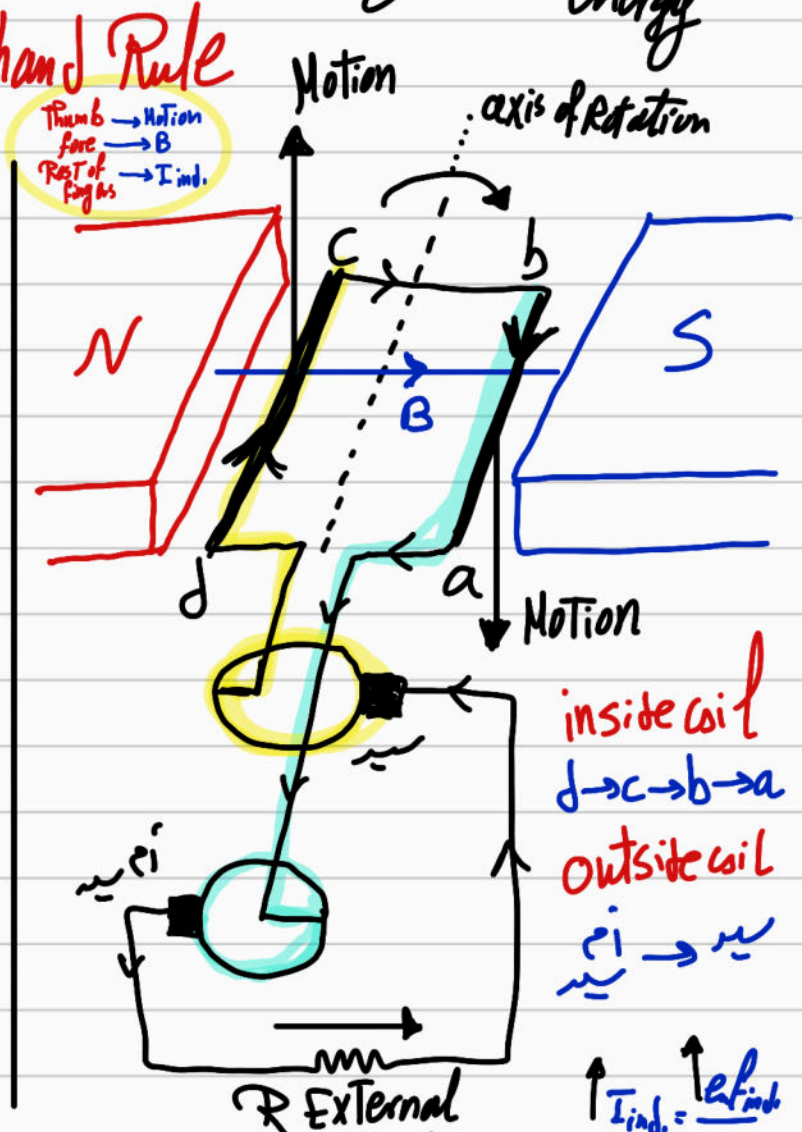
#Sci. idea: Electromagnetic induction

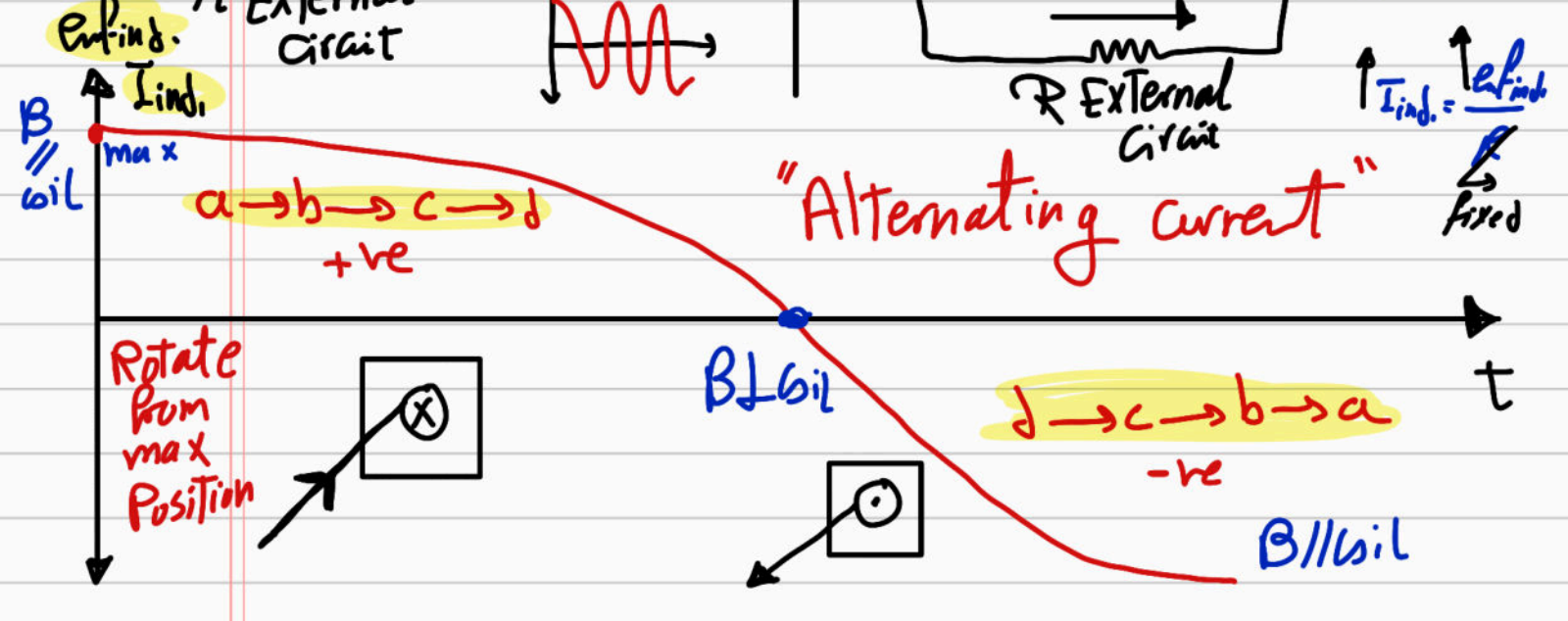
Rotation → Electricity

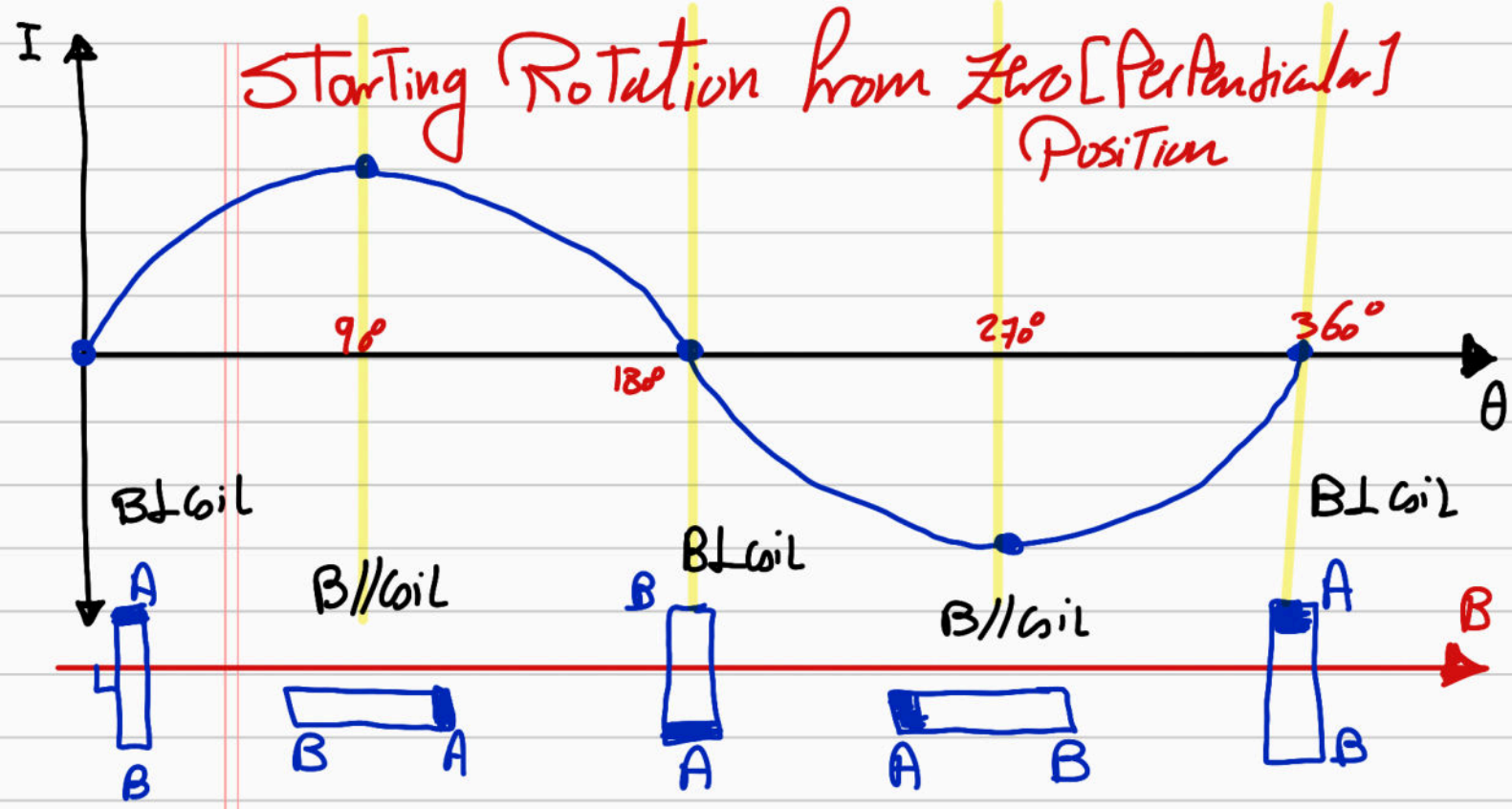
#use: ConVerT Mechanical [Kinetic] energy into Eletric energy

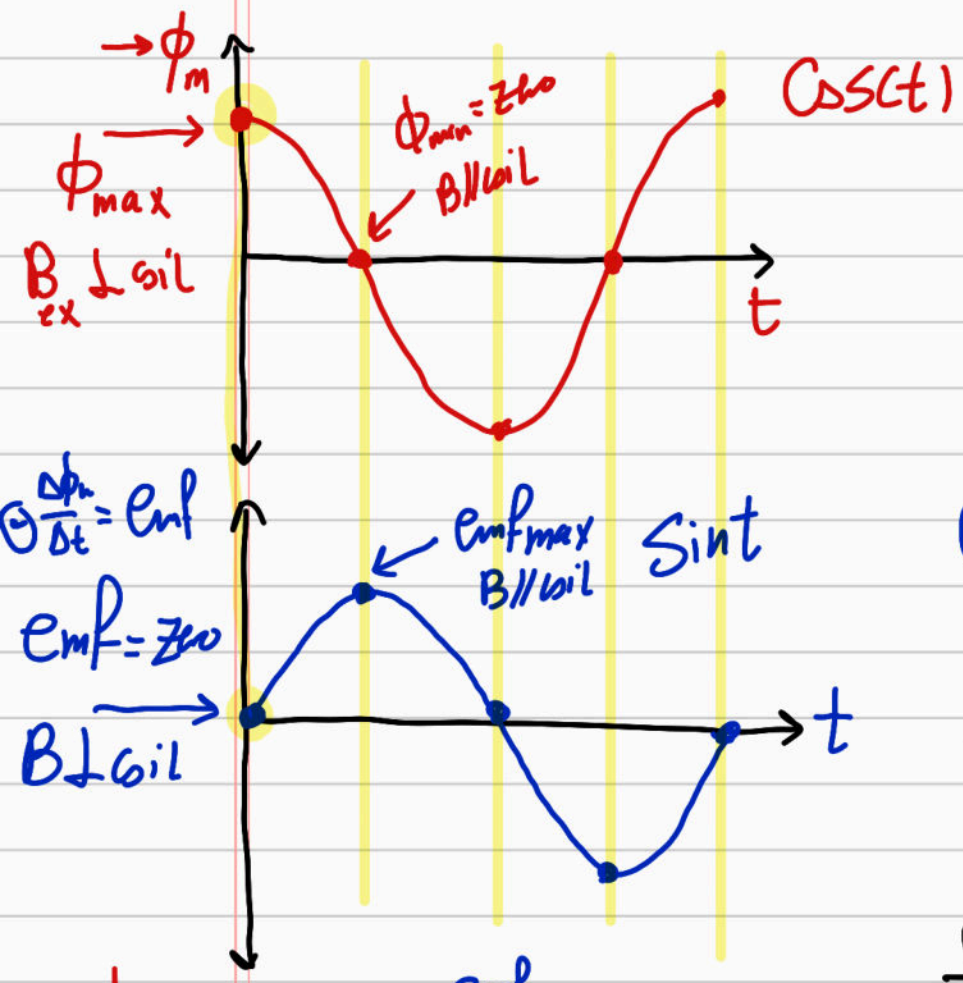
#Design: Fleming Right hand Rule

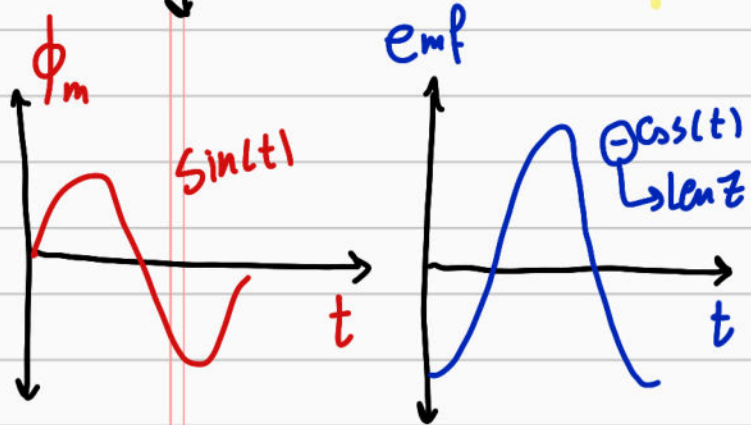




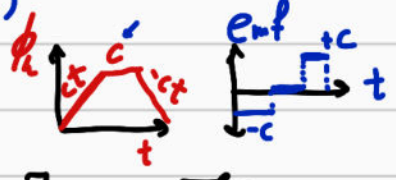

$$\frac{d}{dt}[\cos(t)] = -\sin t$$

$$emf = \Theta N \frac{d\phi_m}{dt}$$

Lenz

$\Theta = \omega t$

$\phi_m$ = Sin(t)

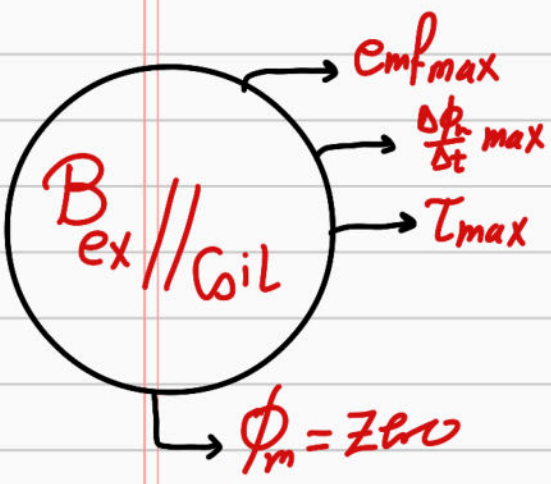


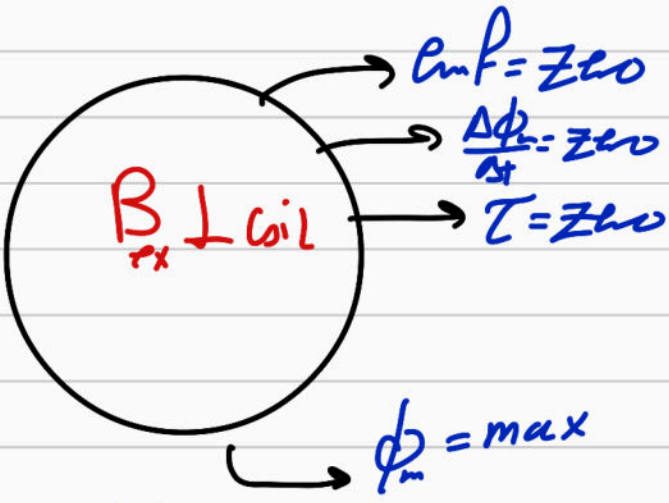

$$\frac{d}{dt}[c] = \text{Zero}$$

$$\frac{d}{dt}[ct] = c$$

$$\frac{d}{dt}[\sin t] = \cos t$$

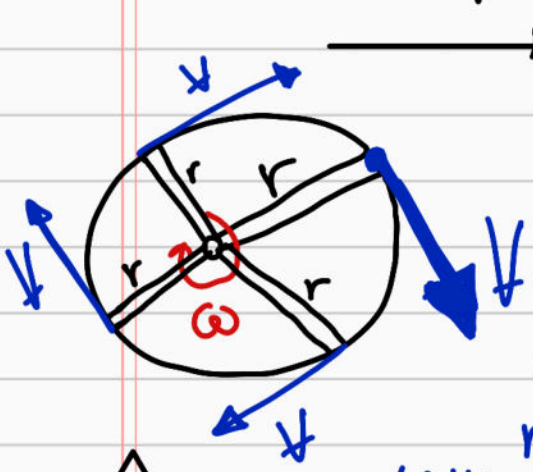
| Positions | $\phi_m$ | $\frac{\Delta\phi_m}{\Delta t}$ | emf | $\tau$ |
|---|---|---|---|---|
| $B_{ex}$ // coil | Zero | max | max | max |
| $B_{ex}$ ⊥ coil | max | Zero | Zero | Zero |

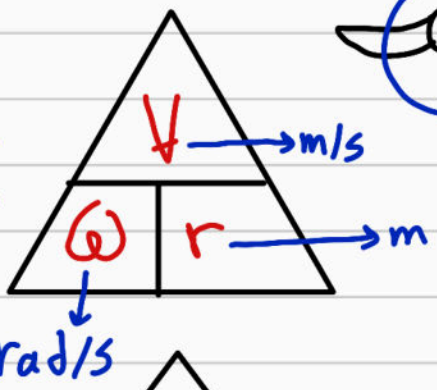



# Note: Linear Speed Vs angular speed/velocity

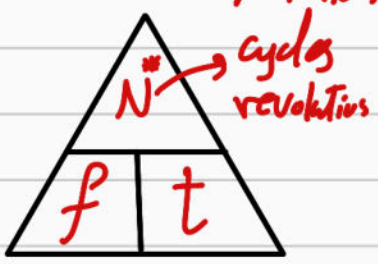
سرعة خطية — سرعة دورانية

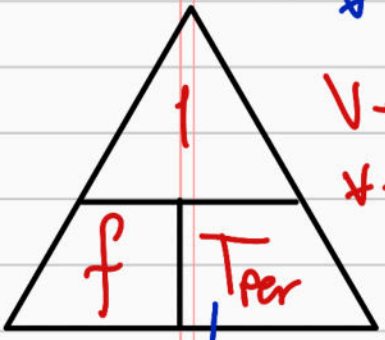




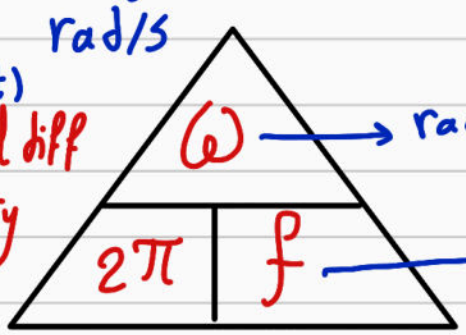




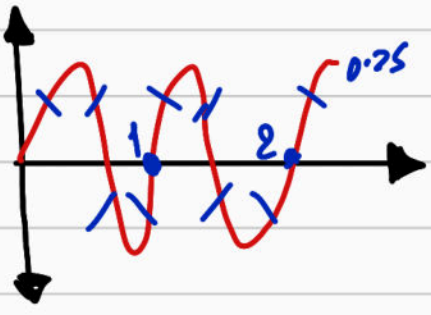

Time To make one complete cycle

(Volt)
V → Potential diff
V → velocity (m/s)




N → no. of Turns of the coil

N* → no. of cycles Rotations Revolution



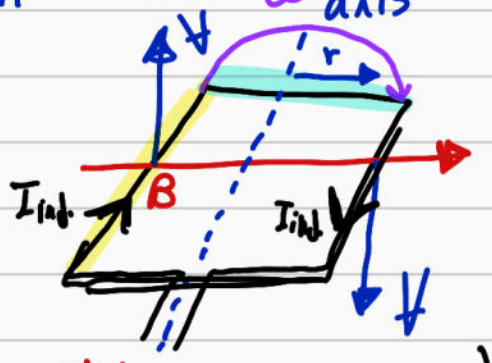
# How To find $emf_{max}$?? $B_{ex} \parallel$ coil

$$emf_{max} = 2 \times emf_{wire}$$

$$= 2 B_{ex} l v N$$

$v$ = linear velocity [m/s]

$l \parallel$ axis of Rotation




$v = \omega r$

$$= 2 B_{ex} l \omega r N$$




$r = \frac{1}{2}$ Width

$$= 2 B_{ex} l \omega \times \frac{1}{2} \text{Width} \times N$$

$A = L \times W$

$$= B_{ex} \times l \times \text{Width} \times \omega N$$

$$emf_{max} = B_{ex} A \omega N$$

Length $\parallel$ axis

Width $\perp$ axis

$$emf_{max} = 2 B_{ex} l v N = B_{ex} A \omega N = I_{max} R$$

By default → Length > Width

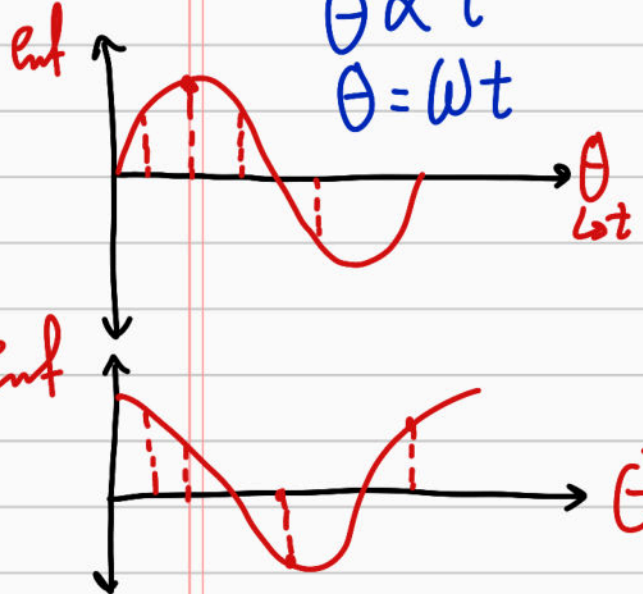
$\theta = \omega t$

$\theta = 2\pi f t$

$\theta = 360^\circ f t$

$\theta \propto t$

$\theta = \omega t$




$$emf_{insta} = emf_{max} \sin\theta$$

$$= 2 B l v N \sin\theta$$

$v = \omega r$

$$emf_{insta.} = emf_{max} \cos\theta$$

$$= B A \omega N \cos\theta$$

$\omega = 2\pi f$

$$\frac{\theta_1}{\theta_2} = \frac{t_1}{t_2}$$

$$emf_{max} = 2Blv \qquad N \rightarrow N = \frac{L_{wire}}{Perimeter} = \frac{L_{wire}}{2\pi r}$$

$v = \omega r$ ; $r = \frac{1}{2}$ Width

$\omega = 2\pi f$ ; 3.14

$f = \frac{N^*}{t}$

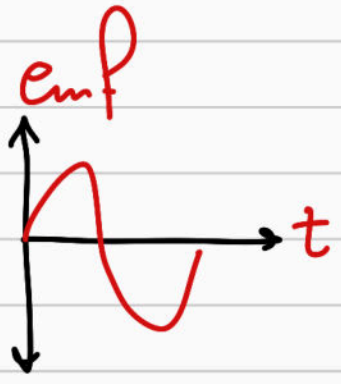
$\theta \propto t \rightarrow \theta = \omega t$ ; $2\pi f = 360°f$ ; $\pi = 180°$

$$emf_{insta.} = emf_{max} \{ \sin\theta \; / \; \cos\theta \}$$

$$emf_{insta.} = BA\omega N \times \sin\theta \quad (\theta = \omega t)$$

$$= BA \times 2\pi \times f \times N \times \sin[2\pi f t]$$

$\frac{22}{7}$ ; 3.14 ; 180° degree




$\pi^{rad} = 180°$

$2\pi^{rad} = 360°$

3.14 ; $\frac{22}{7}$

$\pi \cos[\pi]$

$\rightarrow 180°$

# Rules Summary




$B_{ex} // coil$

$\rightarrow B$

$B_{ex} \perp Normal$

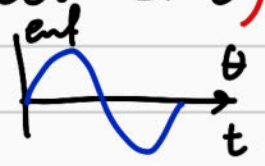
**emf max**

- $2 B_{ex} l v N$
- $B_{ex} A \omega N$
- $I_{max} R$

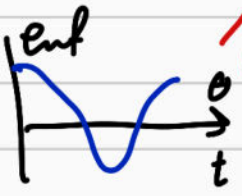
**Torque**

**emf insta.**

- $BA\omega N \sin\theta$ (ex)
  - $B_{ex}$ θ Normal
  - Start Rotation from Zero [Perpendicular]




- $BA\omega N \cos\theta$ (ex)
  - $B_{ex}$ θ coil
  - Start Rotation from parallel max position




**ω**

- $\frac{v}{r}$ → linear speed
- $2\pi f$ → ω → emfmax
- $360^\circ f$ → θ → sin θ, cos θ

$$emf_{insta.} = emf_{max} \sin\theta$$

$$= BA\omega N \times \sin(\omega t)$$

$$= BA\,2\pi f N \times \sin(2\pi f t)$$

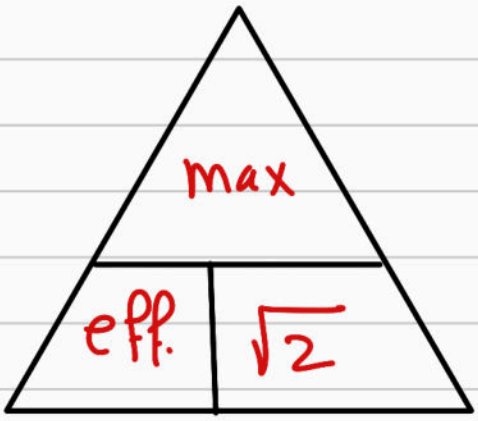
($2\pi$ ↓ 3.14; $2\pi$ ↓ 180°)

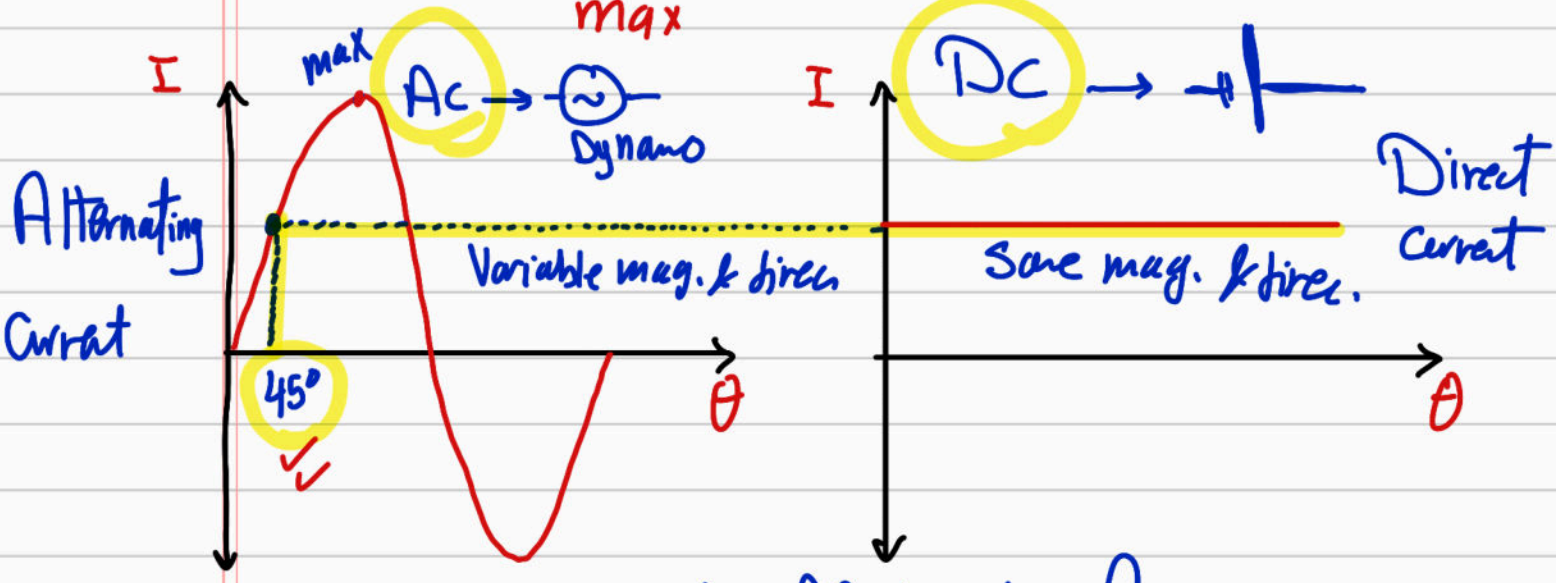
# Effective Value of emf

The Value of AC which produces the same thermal effect as DC at same Time.

$$emf_{eff.} = emf_{max} \times \sin(45)$$

$$= emf_{max} \times \cos(45)$$




Same Thermal effect at the same Time

AC DC

1. max → X
2. eff → same thermal effect
3. Av → X
4. Insta. → X

$$\text{Power (AC)} = emf_{eff} \times I_{eff}$$

$$= \frac{emf_{eff}^2}{R}$$

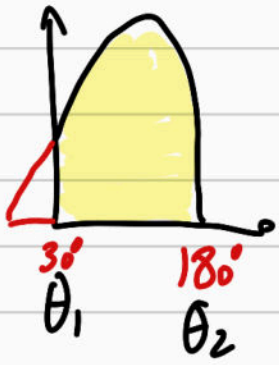
$$= I_{eff}^2 R$$

$$emf_{av} = -N \frac{\Delta \phi_m}{\Delta t}$$

average

| Rotation | Start from Parallel ↙ $\phi_m \to$ Zero | Start from Perpendicular ↙ $\phi_m \to$ max |
|---|---|---|
| Quarter cycle $\times \frac{1}{4}$ $90^0$ | $emf_{av} = -\frac{NAB}{t}[\sin(90)-\sin(0)]$ $= -\frac{NAB\times4}{t\times4} = \frac{-4BAN}{T_{per}}$ direction $emf_{average} = -4B_{ex}ANf$ | $emf_{av} = -\frac{NAB}{t}[\cos(90)-\cos(0)]$ $= \frac{NAB\times4}{t\times4} = \frac{4BAN}{T_{per}}$ $emf_{average} = 4B_{ex}ANf$ |
| half cycle $\frac{1}{2}\times 180^0$ Reverses Turned back upside down | $emf_{av} = -\frac{NAB}{t}[\sin(180)-\sin(0)]$ $= Zero$ ✱ No variation No Induction | $emf_{av} = -\frac{NAB}{t}[\cos(180)-\cos(0)]$ $= \frac{2NAB\times2}{t\times2}$ $emf_{average} = 4B_{ex}ANf$ ✓ |
| three quaters $3/4$ | $emf_{av} = -\frac{NAB}{t}[\sin(270)-\sin(0)]$ $= \frac{NAB\times4/3}{t\times4/3} = \frac{4/3BAN}{T_{per}}$ $= 4/3B_{ex}ANf$ | $emf_{av} = -\frac{NAB}{t}[\cos(270)-\cos(0)]$ $= \frac{NAB^{\times4/3}}{t^{\times4/3}}$ $= 4/3BANf$ |
| Complete cycle | Zero | Zero |
| Generally | $emf_{av} = -\frac{NAB}{t}[\sin\theta_2-\sin\theta_1]$ | $emf_{av} = -\frac{NAB}{t}[\cos\theta_2-\cos\theta_1]$ |
| Using integration | $emf_{av} = \int_{\theta_1}^{\theta_2} \frac{emf_{max}\cos x\,dx}{\theta_2-\theta_1}$ | $emf_{av} = \int_{\theta_1}^{\theta_2} \frac{emf_{max}\sin(x)\,dx}{\theta_2-\theta_1}$ |

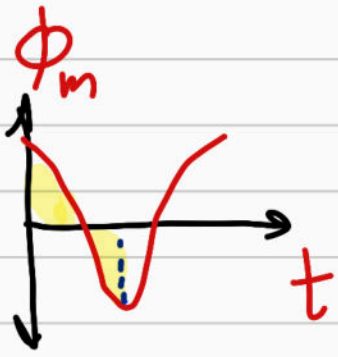
# Average & Max:




$emf_{average} \rightleftharpoons emf_{max}$

$emf_{average} = ??$ (perpendicular ½)

$emf_{max} = 100V$




## ① First solution:

$$\frac{emf_{max}}{emf_{av\,1/2}} = \frac{BA2\pi fN}{4BANf} = \frac{\pi}{2}$$

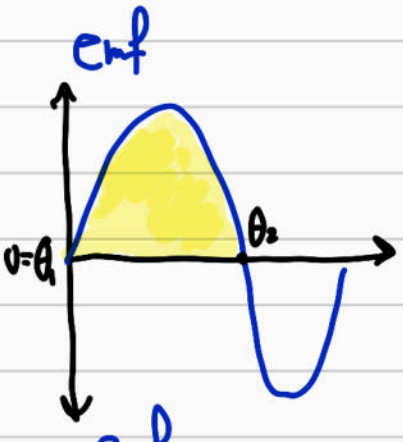
Perpendicular → $\Phi_{max}$, $emf = 0$

$$\frac{100}{emf_{av}} = \frac{\pi}{2} \rightarrow emf_{av} = \frac{100 \times 2}{\pi} = 63.6V$$

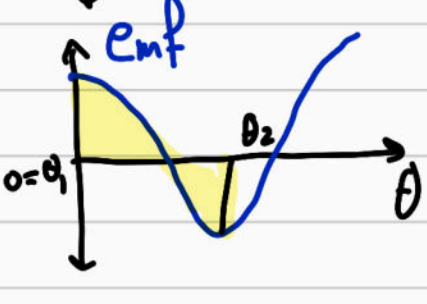
## ② Second solution:

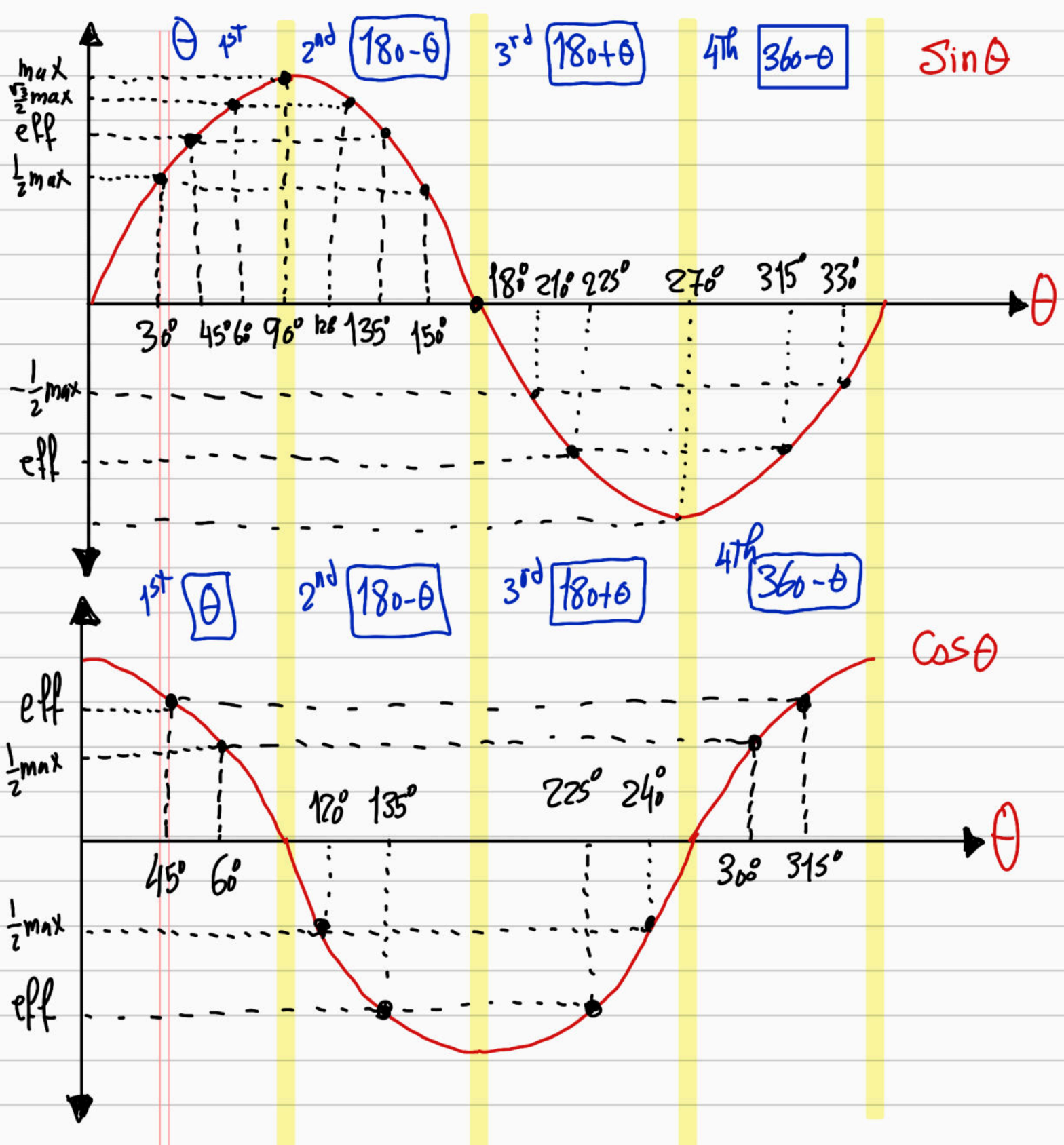
$$emf_{av} = \int_{\theta_1}^{\theta_2} \frac{emf_{max} \times \sin\theta \, d\theta}{\Delta\theta}$$

$$emf_{av\,1/2} = \int_{0}^{180^\circ} \frac{100 \times \sin(x)\, dx}{180 - 0}$$

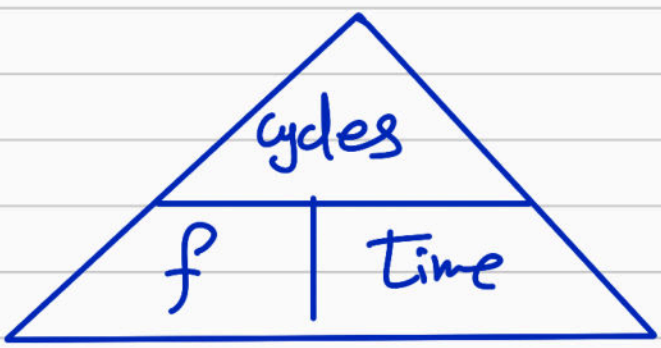


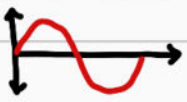
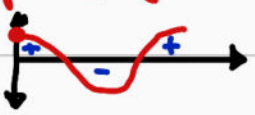

$$emf_{av\,1/2} = \int_{0}^{180} \frac{100 \times \cos(x)\, dx}{180 - 0}$$

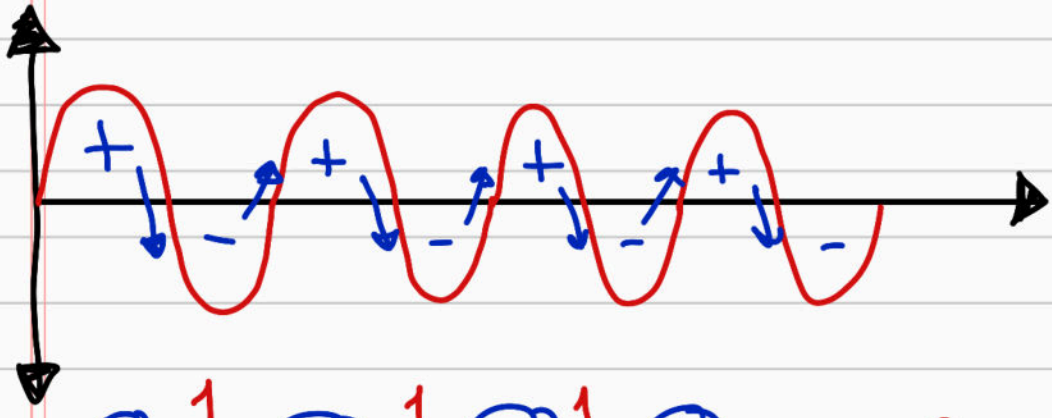


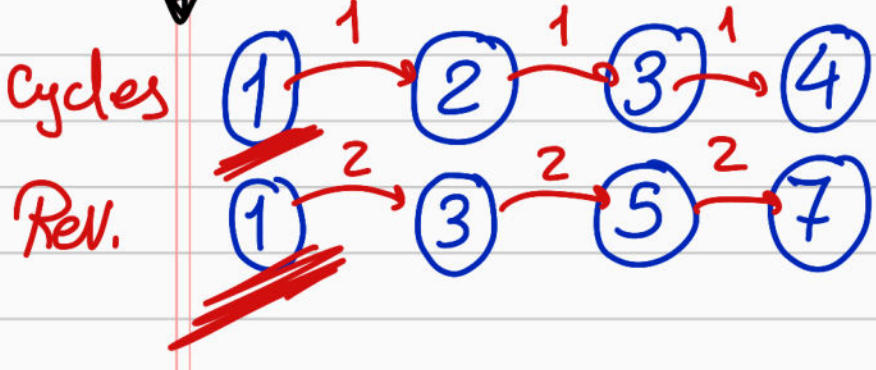

θ 1st
2nd 180-θ
3rd 180+θ
4th 360-θ
Sin θ
max
√3/2 max
eff
1/2 max
30° 45° 60° 90° 120° 135° 150°
180° 210° 225° 270° 315° 330°
θ
-1/2 max
eff


1st θ
2nd 180-θ
3rd 180+θ
4th 360-θ
Cos θ
eff
1/2 max
120° 135°
225° 240°
45° 60°
300° 315°
θ
1/2 max
eff


cycles
f
time

| No. of Times To Reach | Starting From Perfect [Zero] Position  | Starting From parallel [max] position  |
|---|---|---|
| max | 2ft | 2ft +1 |
| Zero | 2ft +1 | 2ft |
| effective | 4ft | 4ft |
| any Value ½ max | 4ft | 4ft |
| Reverses its direction | 2ft -1 | 2ft |

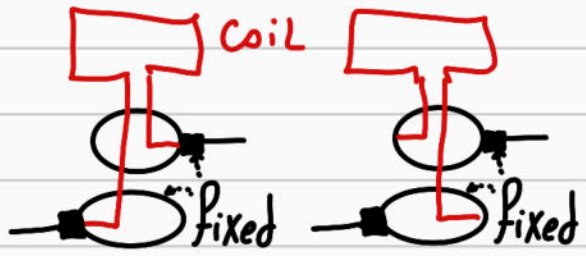
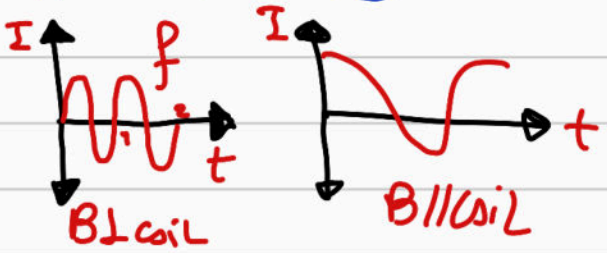
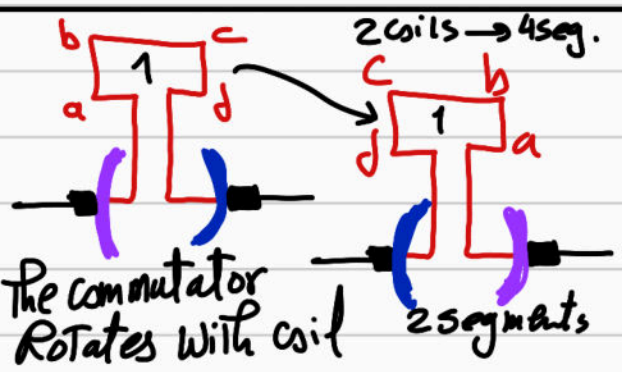
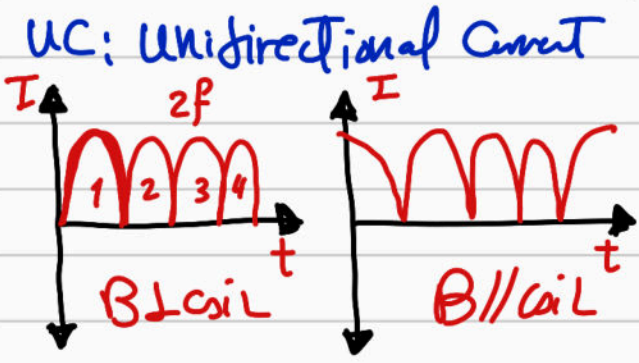
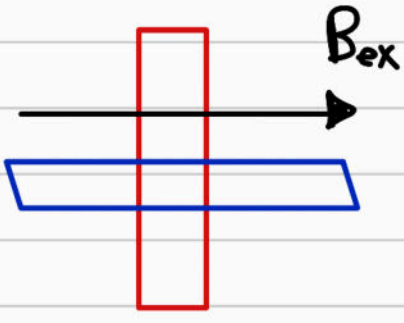
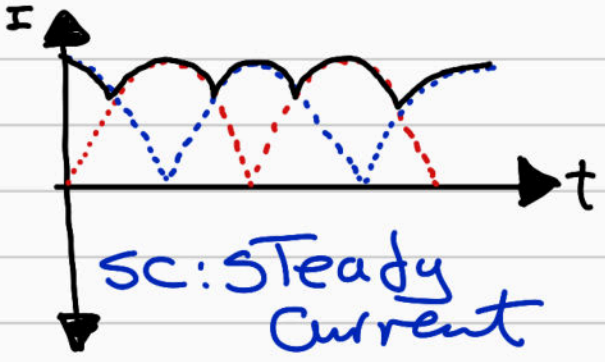
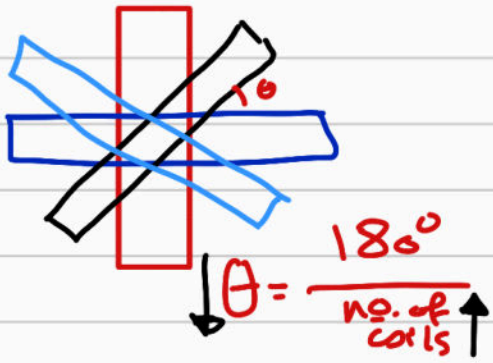
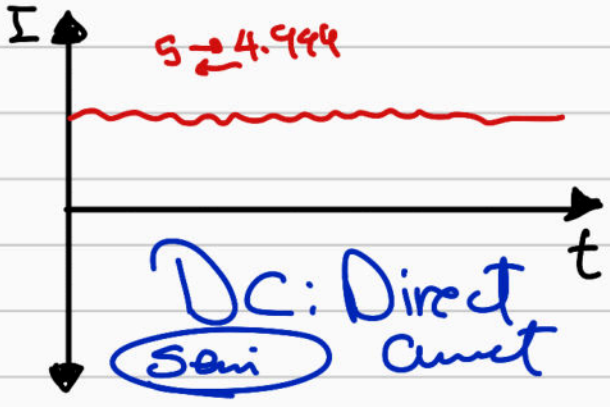


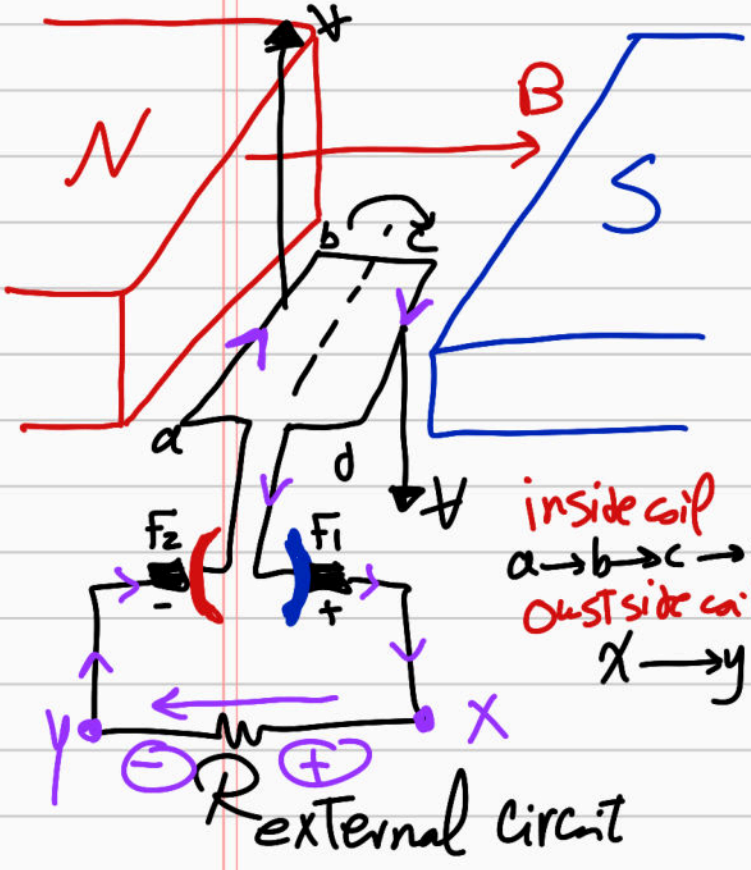



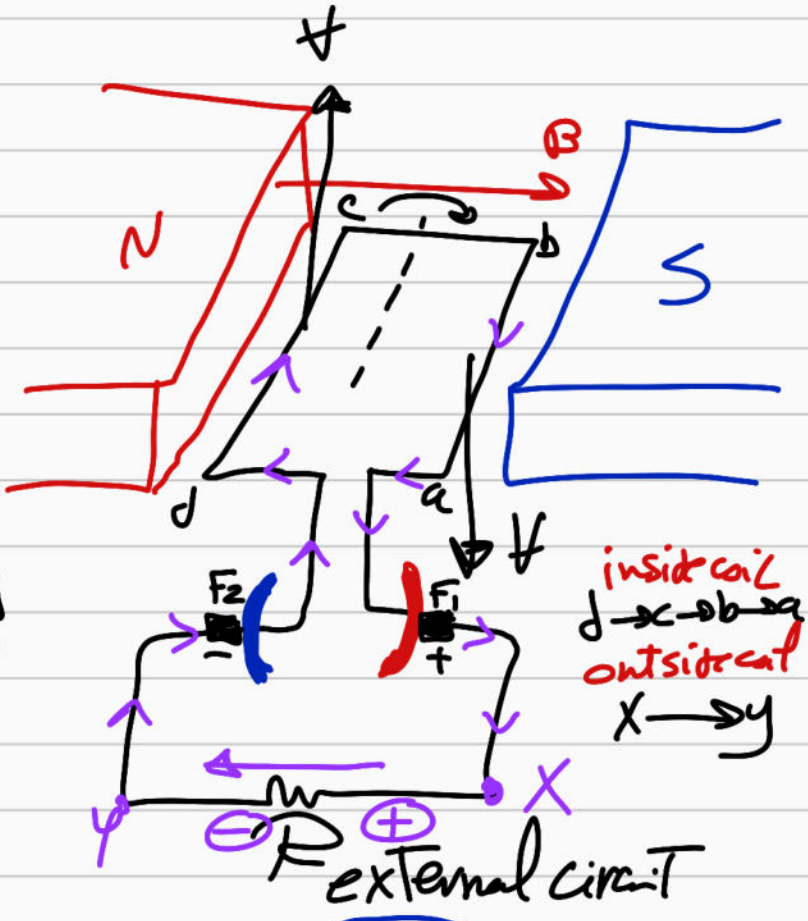

2ft -1

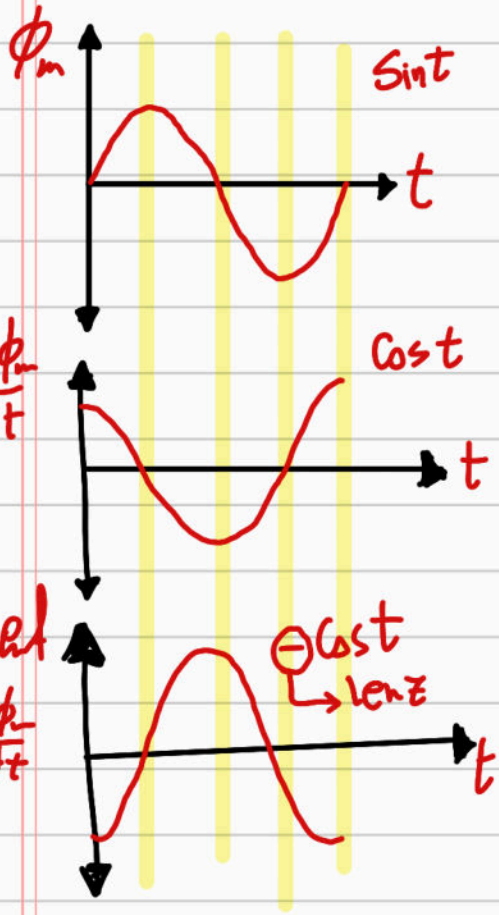
# Rectification

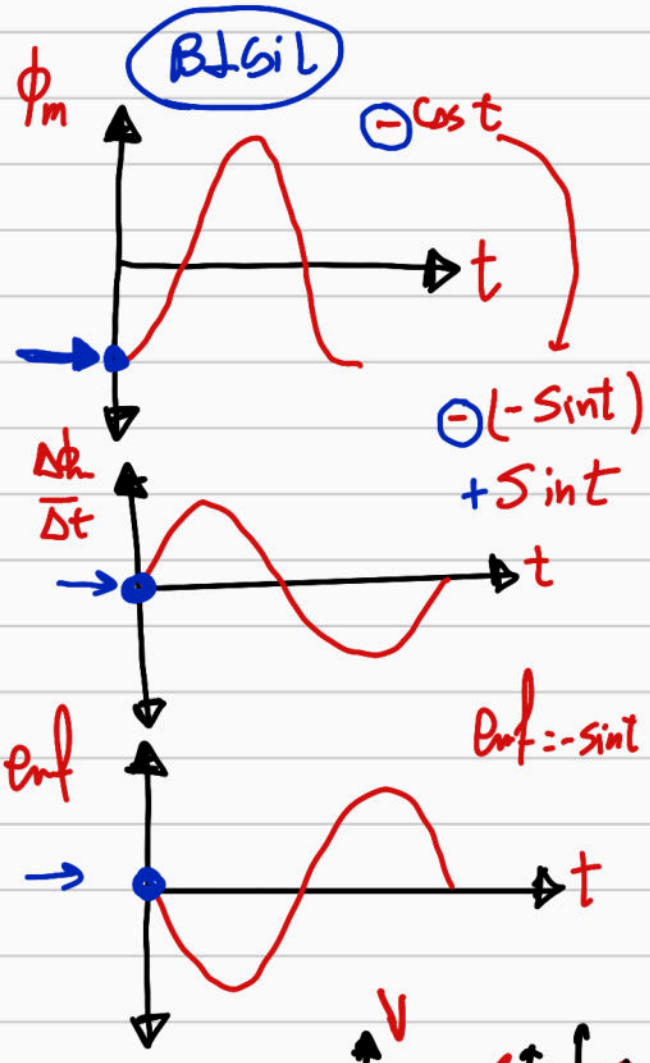
## Ac Dynamo
## Dc Dynamo
external circuit

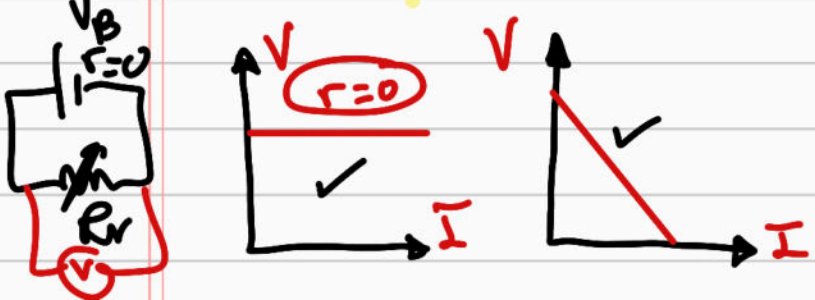
| Modification | Shape | Current in external circuit |
|---|---|---|
| Slip Rings |   | Ac: Alternating current   |
| Split Rings "Commutator" |   | Uc: Unidirectional Current   |
| increase no. of coils |   |   |
| decrease angle between coils |   |   |
| Battery |  |   |


N
S
B
a
b
c
d
F2
F1
inside coil
a→b→c→d
outside coil
x→y
x
y
R
external circuit


N
S
B
c
b
d
a
F2
F1
inside coil
d→c→b→a
outside coil
x→y
x
y
R
external circuit


$\phi_m$
Sint
t
$\frac{d\phi_m}{dt} \to \frac{\Delta\phi_m}{\Delta t}$
Cost
t
emf
⊖Cost
Lenz
t
emf = ⊖N $\frac{d\phi_m}{dt}$


B⊥Sil
$\phi_m$
⊖Cost
t
⊖(-Sint)
+Sint
$\frac{\Delta\phi_m}{\Delta t}$
t
emf
emf = -sint
t


$V_B$
r=0
$R_V$
V
V
r=0
I
V
I
$\downarrow V = V_B - \uparrow I r$
V
$V = I R$
ohm's law
I

2023

# *Transformer

P.C → Variation → Consume

S.C. → Induction → Produce

#Sci. Idea: Mutual induction

#Uses:

**Step up** *

$V_p < V_s$

$N_p < N_s$

$I_p > I_s$

input output

**Step down** *

$V_p > V_s$

$N_p > N_s$

$I_p < I_s$

in out

$P_w = VI$

$\mathcal{E}_{mf_s} = -N_s \frac{\Delta \phi_{m_s}}{\Delta t}$

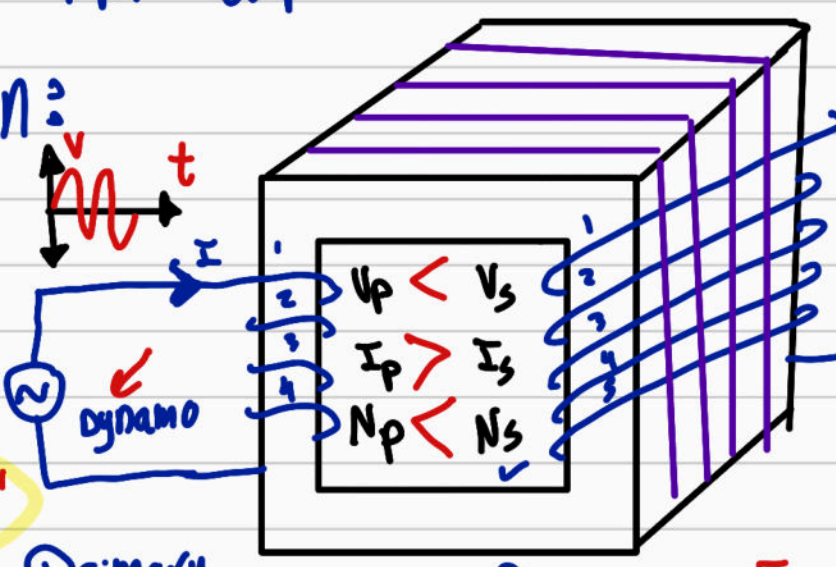
$\mathcal{E}_{mf_p} = -N_p \frac{\Delta \phi_{m_p}}{\Delta t}$

$\frac{\Delta \phi_{m_p}}{\Delta t} = \frac{\Delta \phi_{m_s}}{\Delta t}$

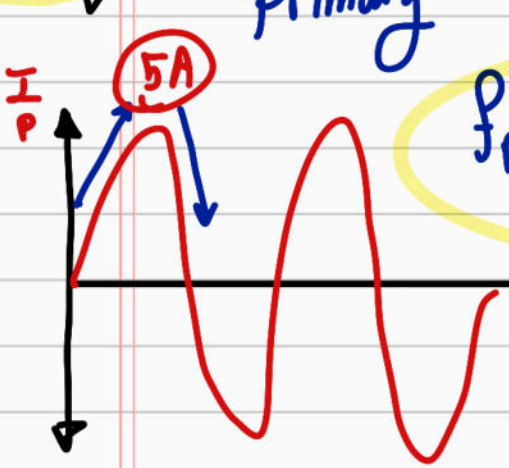
$\frac{\mathcal{E}_{mf_s}}{\mathcal{E}_{mf_p}} = \frac{N_s}{N_p}$

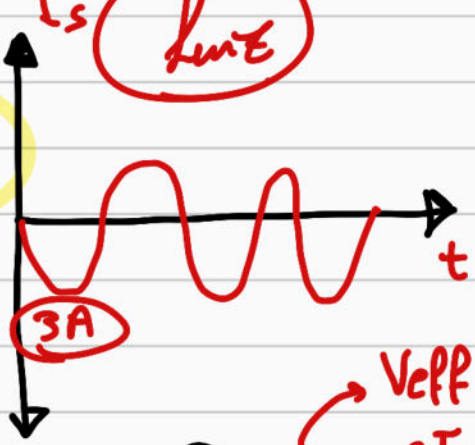
$V_s = I_s$ ~~$P_s$~~ device

#Design:




$f_{primary} = f_{secondary}$




#Rules:

"Ideal Transformer"

$\eta = 100\%$ ضیالی

Theory

1. $P_{w_p} = P_{w_s}$

$P_w = VI = I^2R = V^2/R$ (V: Veff, I: Ieff)

2. $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

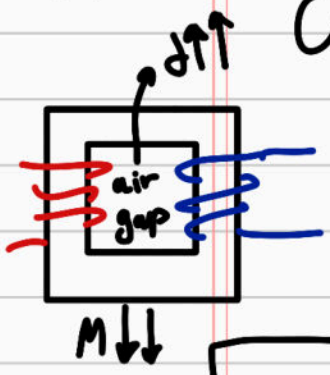
"non-ideal Transformer"

$\eta < 100\%$ ✓

↳ Efficiency of Transformer

1. $P_{wp} > P_{ws}$ 2. $\eta = \frac{P_{ws}}{P_{wp}}$

3. $\eta \frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

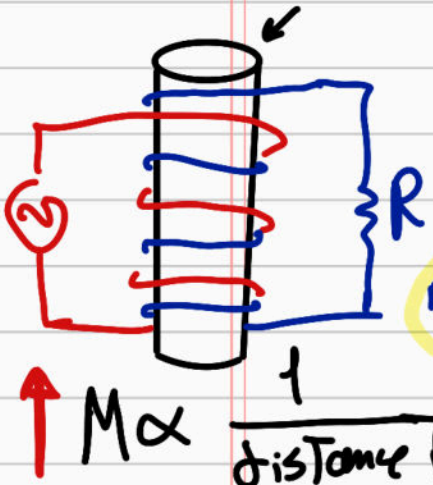



# Power losses insite Transformer

## Mechanical

### Magnetic flux Leakage التسريب

$\frac{\Delta\phi_{leakage}}{\Delta t} = \% \frac{\Delta\phi_{up}}{\Delta t}$




### Magnitization energy

Rearrangment molales of Core "Vibration"

$M = \frac{\mu A N N}{l_s}$

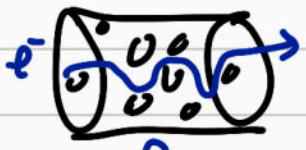
soft Iron Core easily To be Magnitized

$\mu \uparrow$ loss ↓

## Thermal

### Electric Wires

Electrons collite with molales




→ friction heat ↑↑

↳ Pe ↓ heat ↓

Copper

### Eddy Current

Metal Block



Eddy ↑ thermal ↑ loss ↑



# #Notes for solving :

ideal [default]

$P_{wp} = P_{ws}$

$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

$\eta = 100\%$

non-ideal

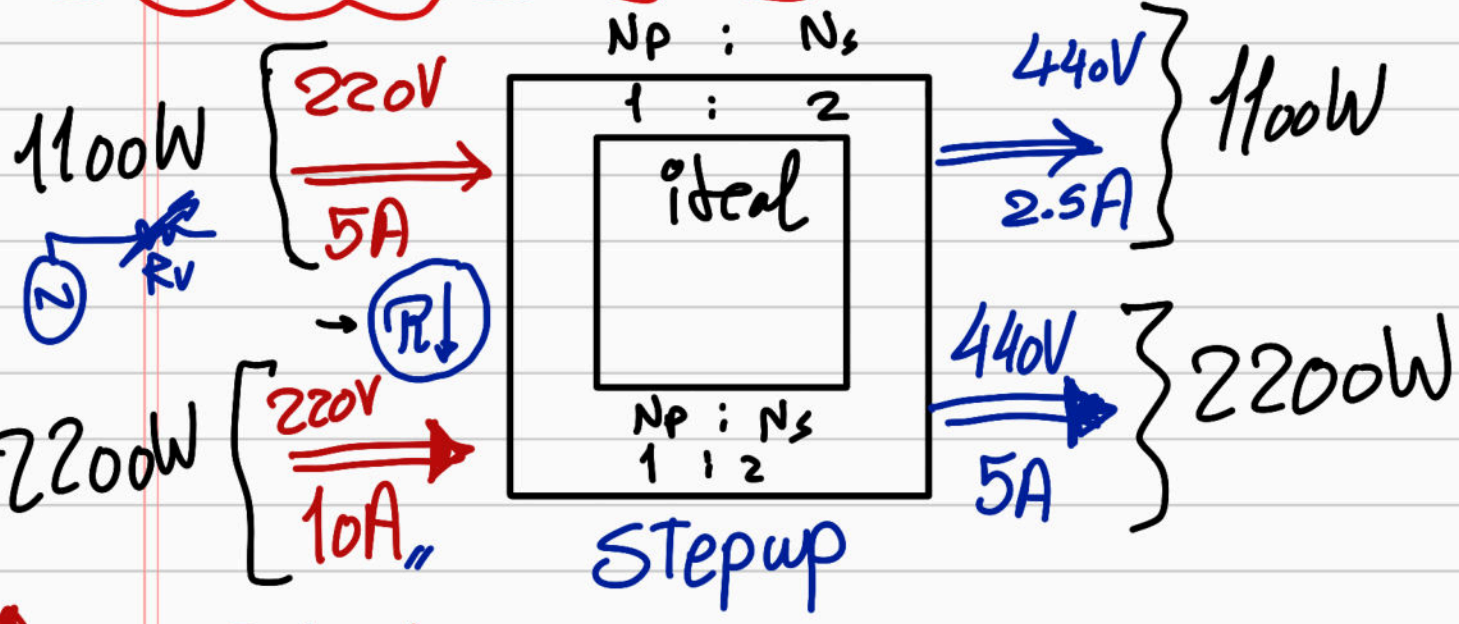
$\eta = \frac{P_{ws}}{P_{wp}} \rightarrow \eta P_{wp} = P_{ws}$

$\eta \frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$

$\eta < 100\%$

$$0.8 = \frac{80}{100} = 80\%$$

solve $\frac{V_p}{V_s} = \frac{N_p}{N_s}$




$P_w = \frac{V^2}{R} \quad 220V$ ; $R \downarrow$

$R = \frac{\rho l}{A}$

NOTE

$V \propto N$

$I \propto \frac{1}{R}$

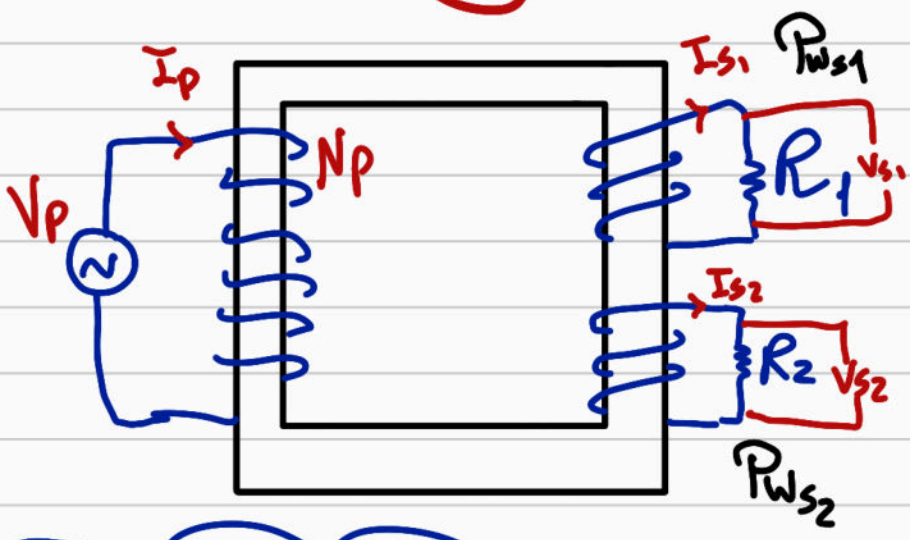
$\eta = \frac{P_{ws}}{P_{wp}} = \frac{V_s I_s}{V_p I_p} \rightarrow \eta \frac{V_s}{V_p} \frac{I_s}{I_p} \rightarrow \eta \frac{V_p}{I_s} = \frac{V_s}{I_p}$

$\eta \frac{V_p}{V_s} = \frac{I_s}{I_p} = \frac{N_p}{N_s}$

# Two or more Secondary coils:

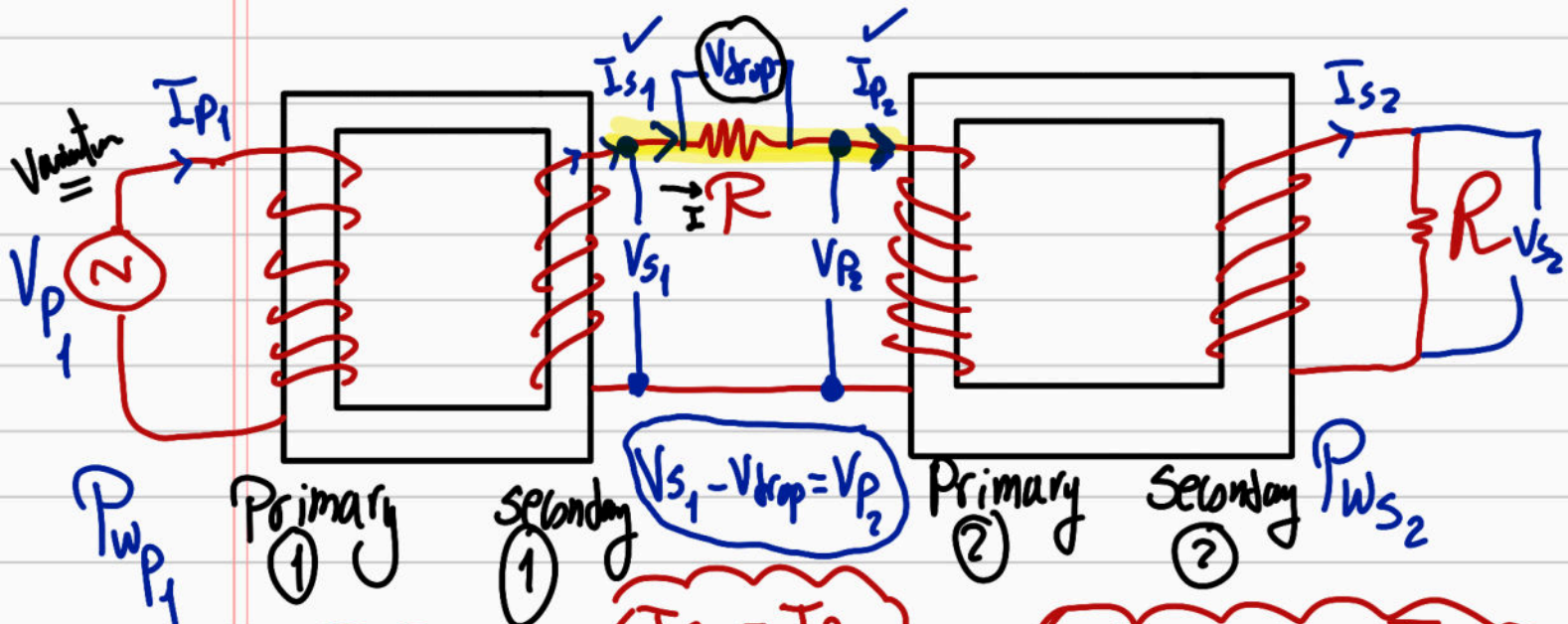
$$\eta \frac{V_p}{V_{s_1}} = \frac{N_p}{N_{s_1}} = \frac{I_{s_1}}{I_p}$$

$$\eta \frac{V_p}{V_{s_2}} = \frac{N_p}{N_{s_2}} = \frac{I_{s_2}}{I_p}$$




$$\eta P_{W_p} = P_{W_{s_1}} + P_{W_{s_2}} + \cdots\cdots$$

# Transformer ⟶ Transformer




$$V_{s_1} - V_{drop} = V_{p_2}$$

$$\frac{V_{p_1}}{V_{s_1}} = \frac{N_{p_1}}{N_{s_1}} = \frac{I_{s_1}}{I_{p_1}}$$

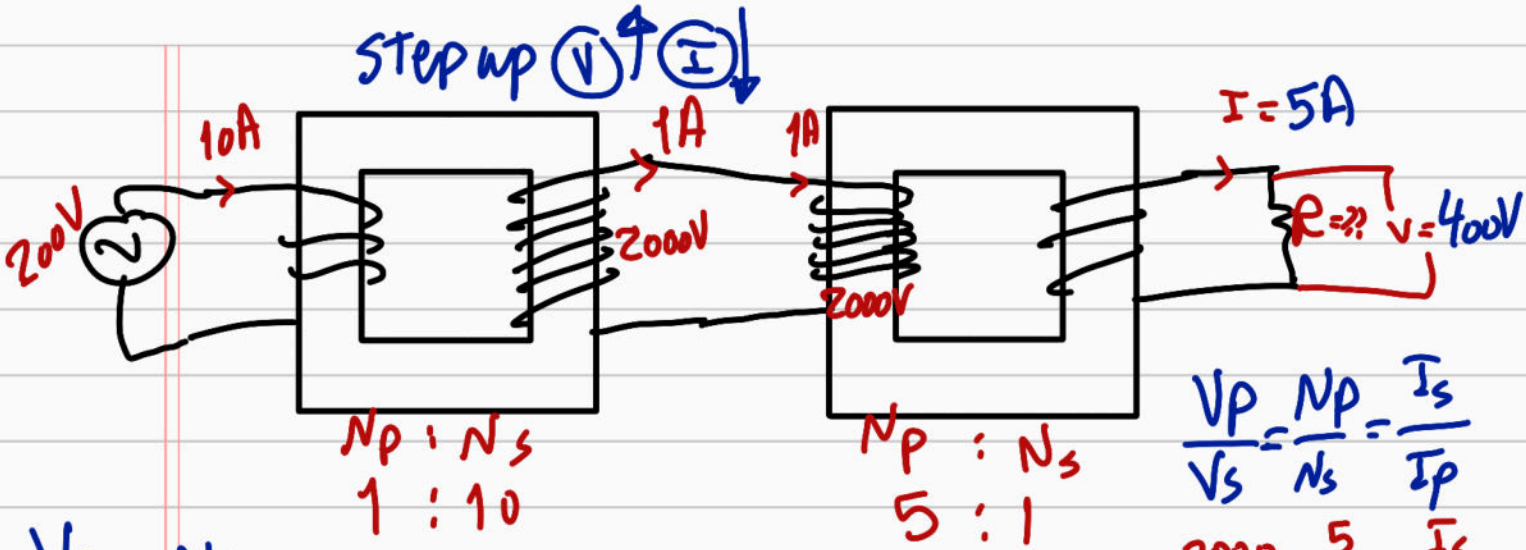
$$I_{s_1} = I_{p_2}$$

$$V_{drop} = I_{s_1} \times R$$

$$P_{W_{loss}} = I_{s_1}^2 \times R$$

$$\frac{V_{p_2}}{V_{s_2}} = \frac{N_{p_2}}{N_{s_2}} = \frac{I_{s_2}}{I_{p_2}}$$

$$P_W = I^2 R$$




$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

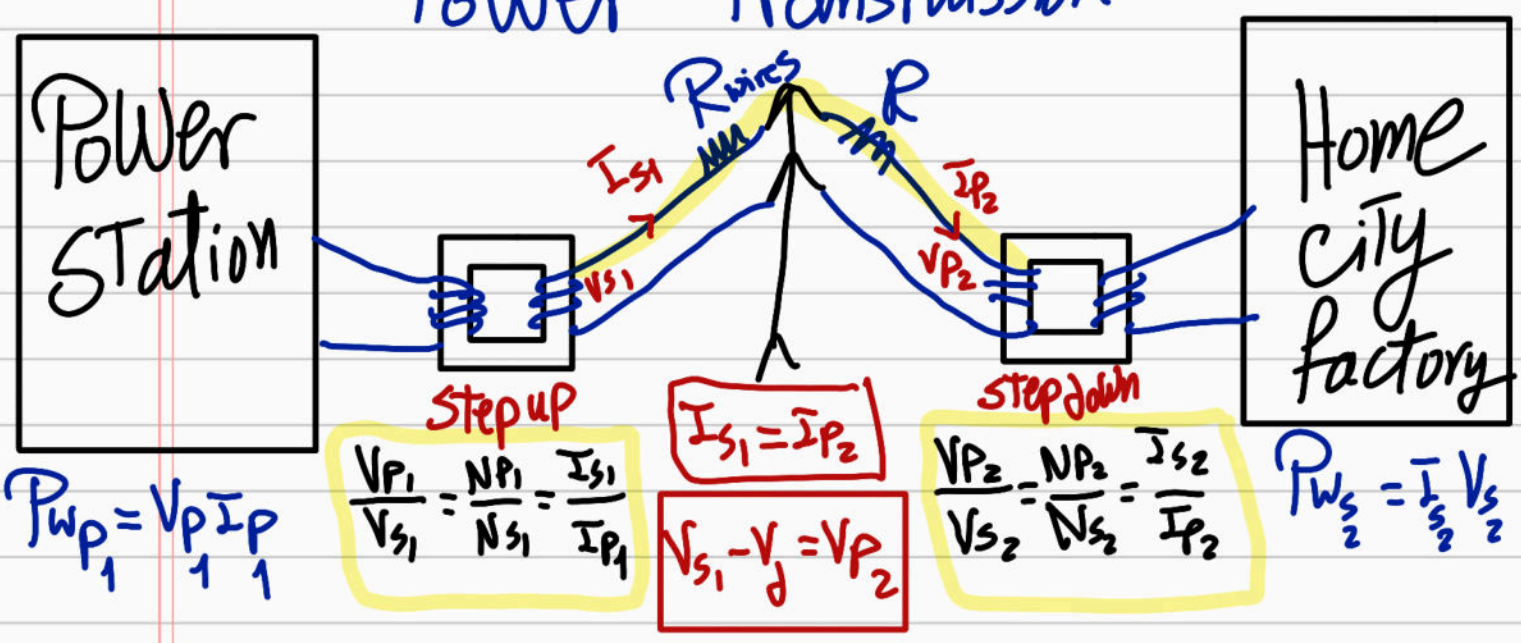
$$\frac{200}{V_s} = \frac{1}{10} = \frac{I_s}{10}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

$$\frac{2000}{V_s} = \frac{5}{1} = \frac{I_s}{1}$$

$$R = \frac{400}{5} = 80\Omega$$

# Power Transmission




$$Pw_{p_1} = V_{p_1} I_{p_1}$$

Step up

$$\frac{V_{p1}}{V_{s1}} = \frac{N_{p1}}{N_{s1}} = \frac{I_{s1}}{I_{p1}}$$

$$I_{s_1} = I_{p_2}$$

$$V_{s_1} - V_j = V_{p_2}$$

Step down

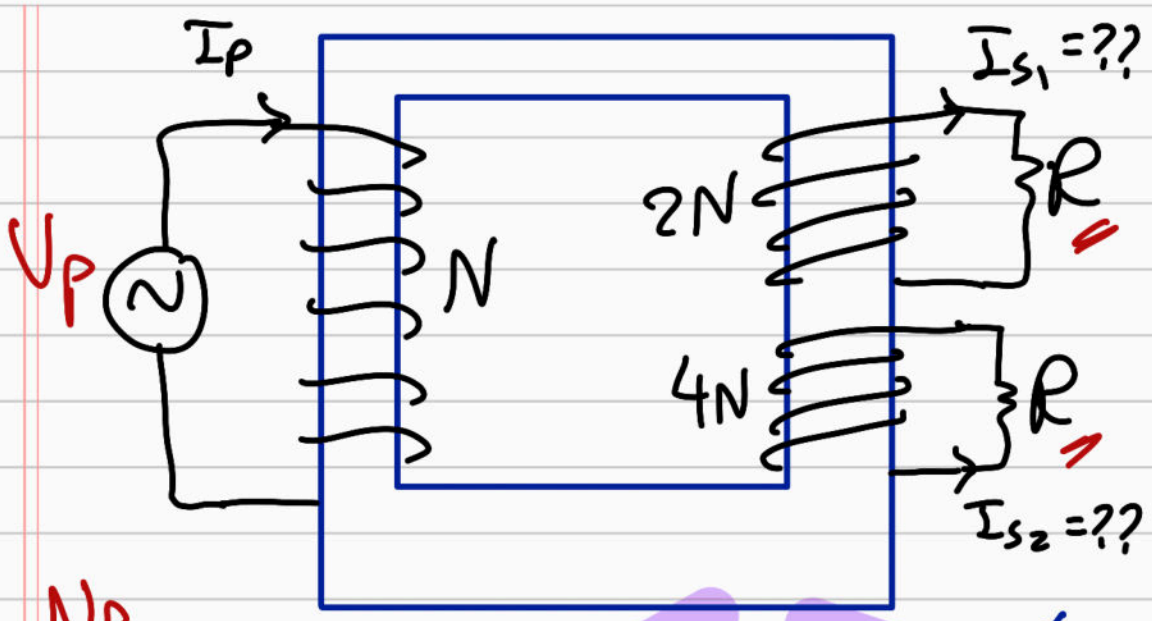
$$\frac{V_{p_2}}{V_{s_2}} = \frac{N_{p_2}}{N_{s_2}} = \frac{I_{s_2}}{I_{p_2}}$$

$$Pw_{s_2} = I_{s_2} V_{s_2}$$

$$V_{drop} = I_{s_1} \times R$$

$$Pw_{loss} = I_{s_1}^2 \times R$$

Q61




$$\frac{V_p}{V_{s_1}} = \frac{N_p}{N_{s_1}}$$

~~$\frac{I_{s_1}}{I_p} = \frac{N_p}{N_{s_1}} \rightarrow \frac{I_{s_1}}{I_p} = \frac{N}{2N} \rightarrow I_{s_1} = \frac{1}{2} I_p$~~

$$\frac{V_p}{V_{s_1}} = \frac{N}{2N} \rightarrow \frac{V_p}{V_{s_1}} = \frac{1}{2} \rightarrow \boxed{V_{s_1} = 2V_p} \#$$

$$\frac{V_p}{V_{s_2}} = \frac{N}{4N} \rightarrow \frac{V_p}{V_{s_2}} = \frac{1}{4} \rightarrow \boxed{V_{s_2} = 4V_p} \#$$

$$\frac{V_{s_1}}{V_{s_2}} = \frac{I_{s_1} \times R}{I_{s_2} \times R} \rightarrow \frac{2V_p}{4V_p} = \frac{I_{s_1}}{I_{s_2}}$$

$$I_{s_2} = 2 I_{s_1}$$

$P_{wp} = V_p I_p$
$P_{ws} = V_s I_s$

$P_{wp} = P_{ws_1} + P_{ws_2}$
$V_p I_p = V_{s_1} I_{s_1} + V_{s_2} I_{s_2}$

$$V_p I_p = 2V_p \times I_{s_1} + 4V_p \times I_{s_2}$$

$$I_p = 2I_{s_1} + 4I_{s_2}$$

$$I_p = 2I_{s_1} + 4(2I_{s_1})$$

$$I_p = 2I_{s_1} + 8I_{s_1}$$

$$I_p = 10 I_{s_1} \rightarrow \boxed{I_{s_1} = \frac{1}{10} I_p}$$

$$I_{s_2} = 2I_{s_1} = 2 \times \frac{1}{10} I_p = \frac{1}{5} I_p$$

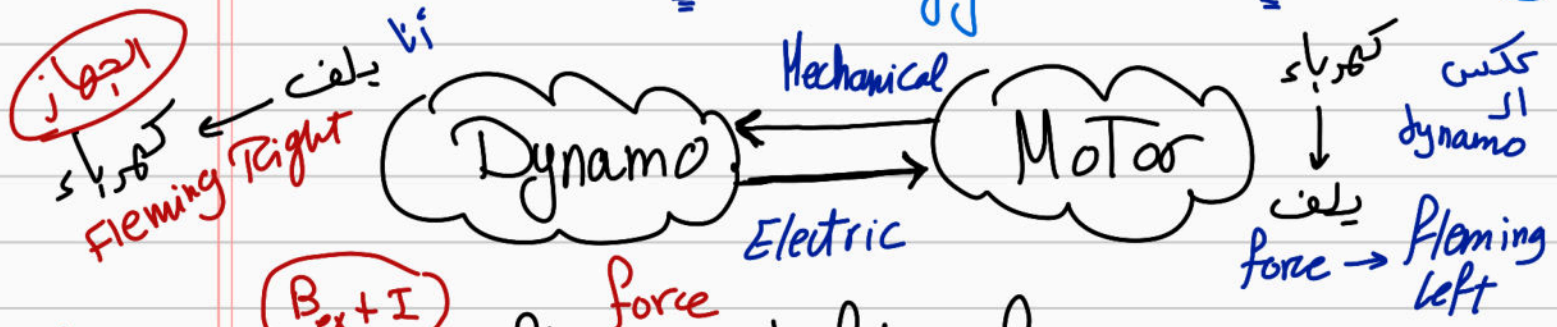
$$\boxed{I_{s_2} = \frac{1}{5} I_p}$$

# Dc Motor

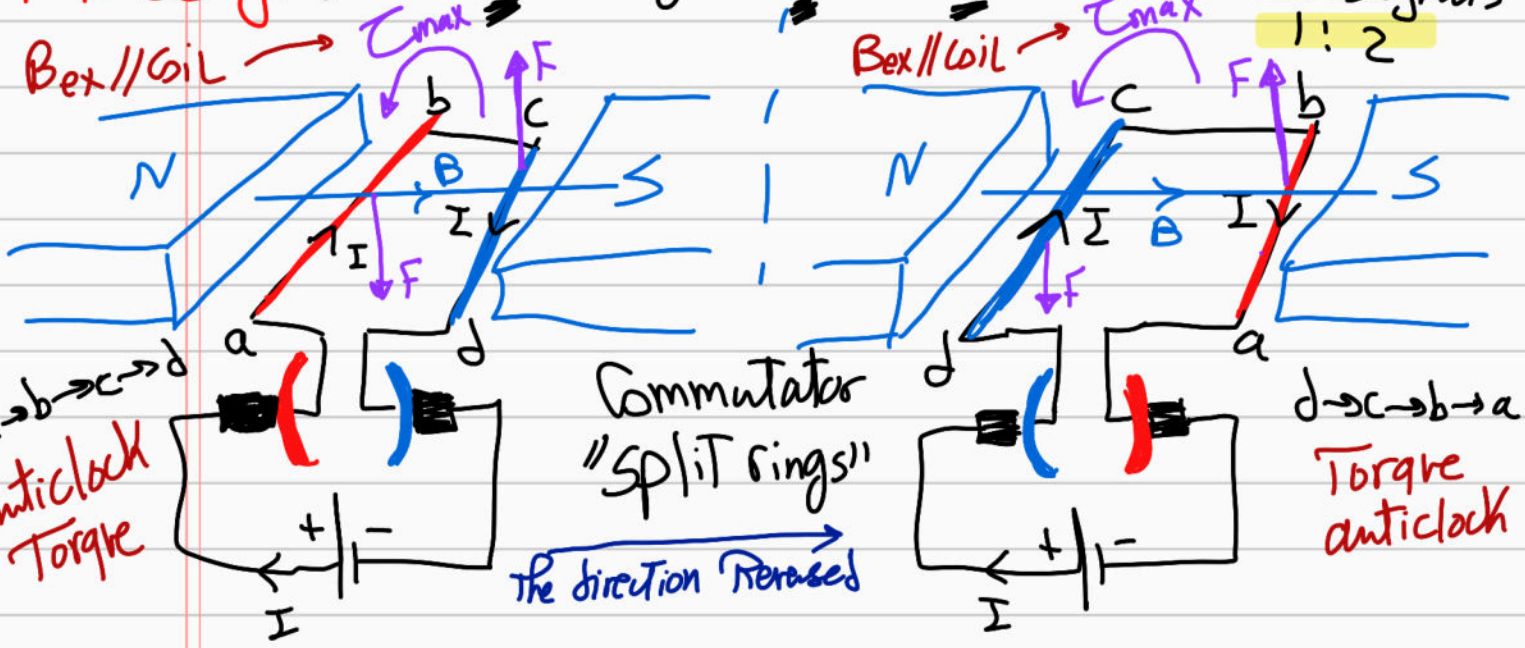
#Sci. Idea: Magnetic Torque → $\tau_{max} = B_{ex} I A N = F \times d$ ← B // Coil Plane




# Uses: Convert Electric energy into Mechanical Energy

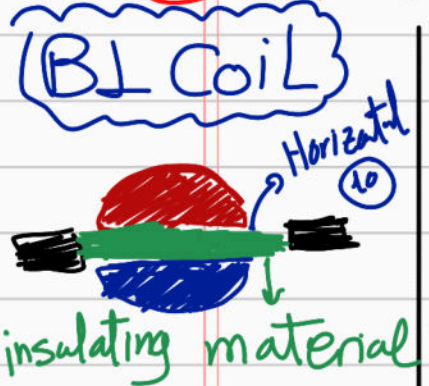



#Design: $B_{ex} + I$ → force Fleming Left force




The current changes its direction once each ½ cycle starting from Zero position [B⊥coil]

* (I) كمان نكس الإتجاه $\tau$ = Zero *




1. I = Zero
2. $\tau$ = Zero
3. $|\vec{m} \times \vec{B}|$ = Zero
4. F = Zero
5. $\phi_{max} = B_{ex} A$
6. the current starts To Reverse its direction
7. Brushes Touching insulating material
8. Velocity ≠ 0
9. inertia continue

1 The coil doesn't stop even at Zero position. [B ⊥ coil] $\tau$ = Zero

↳ due to inertia* → Variation $\Phi_m$ → Induction self

2 The speed of rotation of the coil of the motor keeps its speed constant* even the torque changes. ↑ $emf = -N\frac{\Delta\Phi}{\Delta t}$ ↑ speed ↑

↳ due to back induced emf*

↑ $emf_{back} \propto$ speed ↑

↓ $I = \frac{V_B \ominus emf_{back}}{R}$ ↑

↓ speed $\propto$ ↓ $\tau = B_{ex} I A N$ ↓

direction

self Induction

↑ $emf_{back} = (-) L \frac{\Delta I}{\Delta t}$ ↑

↑ $\frac{\Delta I}{\Delta t}$ ↓ $\propto$ speed ↑

# efficiency of Motor

(1) inc. no. of coils ⇝ Coil : Segments
1 : 2

(2) decrease angle between coils ⇝ $I_{max}$ → $B_{ex} \parallel$ Coil

(3) increase Mechanical Power $\tau = B_{ex} I A N$

(a) inc. external Mag. field

(b) inc. Current ⟶ $\uparrow I = \frac{V_B \uparrow}{R \downarrow}$

(c) inc. Area

(d) inc. no. of Turns

(e) inc. no. of coils

